Research in Computational Biology book

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Contents

Chapter 1

Prerequisites

This is a *sample* book written in **Markdown**. You can use anything that Pandoc's Markdown supports, e.g., a math equation $a^2 + b^2 = c^2$.

The **bookdown** package can be installed from CRAN or Github:

```
install.packages("bookdown")
# or the development version
# devtools::install_github("rstudio/bookdown")
```

Remember each Rmd file contains one and only one chapter, and a chapter is defined by the first-level heading `#`.

To compile this example to PDF, you need XeLaTeX. You are recommended to install TinyTeX (which includes XeLaTeX): https://yihui.org/tinytex/.

Chapter 2

Basics of R

2.1 Basic Steps

- R can be accessed by clicking an icon or entering the command “R” at the system command line, also refered to as Terminal
- This produces a console window or causes R to start up as an interactive program at the current terminal window
- R works fundamentally by a question-and-answer model: enter a line with a command and press Enter
- Then the program does something relevant
- In this course, we will use RStudio and a file format called R Markdown, which allows us to code using not only R, but bash and Python, two other languages we may refer to as the course progresses

2.2 Library ISwR

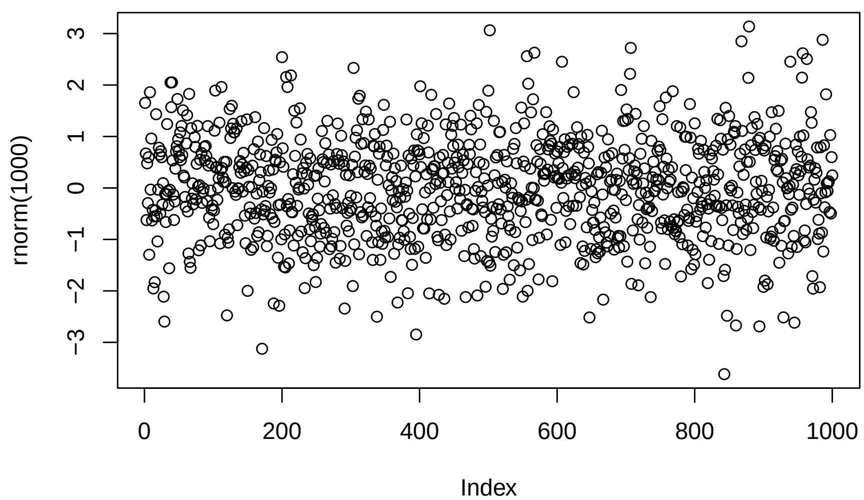
- This notebook used the book Introductory Statistics with R, by Peter Dalgaard as a reference
- For the book, R library ISwR (Introductory Statistics with R) can be freely downloaded
- All examples in the book used as reference should run provided that ISwR library is not only installed by also loaded into the current search path
- Library can be installed and loaded by typing the following command into an R chunk

```
install.packages("ISwR")
library("ISwR")
```

2.3 Plot Random Numbers

- For a first impression of what R can do, let´s try plotting a graph
- Need to insert the R chunk and use the plot function

```
plot(rnorm(1000))
```



2.4 A Potent Calculator

- R can process simple and complex arithmetic expression and produce a result for the user

```
2 + 2
```

```
## [1] 4
```

- R can also be used to do other standard calculations. Here is how to calculate e to the power of -2

```
exp(-2)
```

```
## [1] 0.1353353
```

- Other than the R chunks, these calculations can be made using the RStudio Console
- In RStusio, when using the R Markdown format (.Rmd), we can insert R, Bash (terminal) and Python chunks
- Proceed to illustrate that in the RStudio IDE

2.5 Assignments

- Assignments are made based on the necessity to store the results of calculations and use these results in downstream processing steps in an entire algorithm or pipeline
- Like other languages, R has symbolic **variables**: names that can be used to represent values
- To assign the value 2 to variable x, we can enter the following command

```
x <- 2
```

- The character <- is called the assignment operator
- There is no immediate visible result, but from now on, x has the value 2 and can be used in subsequent arithmetic operations
- We can "ask" R which value is stores in the x variable

```
x
```

```
## [1] 2
```

2.6 Operations after assignment of variable

- Below, our variable x, is used to perform other calculations

```
x
```

```
## [1] 2
```

- Addition

```
x + x
```

```
## [1] 4
```

- Multiplication

```
5*x
```

```
## [1] 10
```

- Potentiation

```
x^3
```

```
## [1] 8
```

2.7 Vectorized Arithmetic

- It is not useful to use one number at a time to run statistics
- One strenght of R is that it can handle entire data vectors as single objects
- A data vector is an array of numbers and a vector variable can be constructed like this

```
weight <- c(60, 72, 57, 90, 95, 72)

## To look at the vector variable, just type its name again
weight
```

```
## [1] 60 72 57 90 95 72
```

- You can do calculations with vectors just like ordinary numbers, so long as they have the same length

- One exception to this rule that we will see will be when we use the mean of weigths of persons (represented by xbar)
- In that case, the mean will be one single number, which will be subtracted from each sample value
- Suppose the weight vector indicates the weight of men in kilograms
- One simple formula to indicate whether a person is obese or not, is the body mass index (BMI)
- BMI is calculated by dividing the person´s weight by the square of their height, in meters
- Therefore, in R, we need to have a vector with the height values to calculate the bmi vector, containing the body-mass index for the individuals indicated in the weight vector

```
height <- c(1.75, 1.80, 1.65, 1.90, 1.74, 1.91)
bmi <- weight/height^2
bmi
```

```
## [1] 19.59184 22.22222 20.93664 24.93075 31.37799 19.73630
```

2.8 Calculate the Mean of the variable weight

- The mean is calculated by the sum of the observations divided by the total number of observations

$\overline{x} = \sum x_i$ /n

- Let´s calculate the mean of the variable weight

```
sum(weight)
```

```
## [1] 446
```

```
sum(weight)/length(weight)
```

```
## [1] 74.33333
```

2.9 Calculate the Standard-Deviation of the variable weight

- Standard-deviation can be calculated with the following equation

$$SD = \sqrt{(\sum (x_i - \overline{x})^2)/(n-1)}$$

- xbar, the mean of variable weight, can be calculated using the sum and length of variable weight

```
xbar <- sum(weight)/length(weight)
xbar
```

```
## [1] 74.33333
```

- Now we can calculate the difference of each replicate in the weight variable and the mean of the weight variable, one by one

```
weight - xbar
```

```
## [1] -14.333333  -2.333333 -17.333333  15.666667  20.666667  -2.333333
```

- Notice how R uses xbar, which has length one, to calculate the new weight - xbar data vector
- xbar is recycled and subracted from every element in the weight variable (weight data vector)

2.10 Calculation of the Standard Deviation

- Calculate the squared deviations

```
(weight - xbar)^2
```

```
## [1] 205.444444   5.444444 300.444444 245.444444 427.111111   5.444444
```

- Calculate the sum of squared deviations

```
sum((weight - xbar)^2)
```

```
## [1] 1189.333
```

Calculate the standard deviation

```
sqrt(
  sum(
    (weight - xbar)^2/
      (length(weight)-1)
)
)
```

```
## [1] 15.42293
```

2.11 Standard Statistical Procedures

- It is a standard medical practice to access whether a person is obese or not using validated scientific criteria
- As a simple procedure to show this concept, let's assume that an individual with a normal weight should have a BMI in the range 20-25
- We want to know if our data deviates from the normal range of BMI
- In R, this can be done using a statistical test called t-test
- You do not need to understand what a t-test is, just remember that is is used to evaluate the distribution of sample values compared to the normal distribution
- You can use a one-sample t-test to assess whether the six persons' BMI can be assumed to have mean 22.5 given that they come from a normal distribution
- You can do that using the function t.test

2.12 Standard Statistical Procedures: t-test

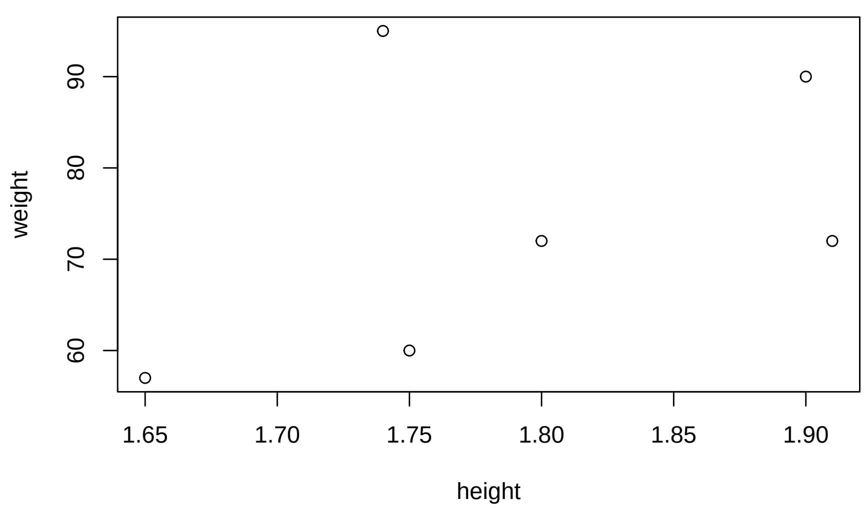
```
t.test(
  bmi,
  mu = 22.5
)
```

```
##
##  One Sample t-test
##
## data:  bmi
## t = 0.34488, df = 5, p-value = 0.7442
## alternative hypothesis: true mean is not equal to 22.5
## 95 percent confidence interval:
##  18.41734 27.84791
## sample estimates:
## mean of x
##  23.13262
```

2.13 Plot Graphics

- Let's now plot a scatterplot of the height and weight of individuals

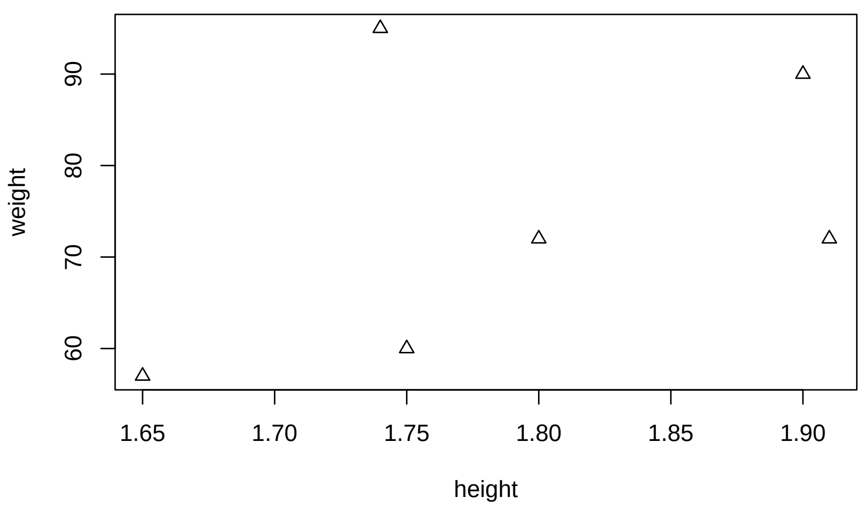
```
plot(height, weight)
```



2.14 Plot Graphics Modifying the plotting character

- You will frequently want to modify drawing of your graphs in various ways
- One way is usig the parameter "plotting character", pch

```
plot(height, weight, pch =2)
```



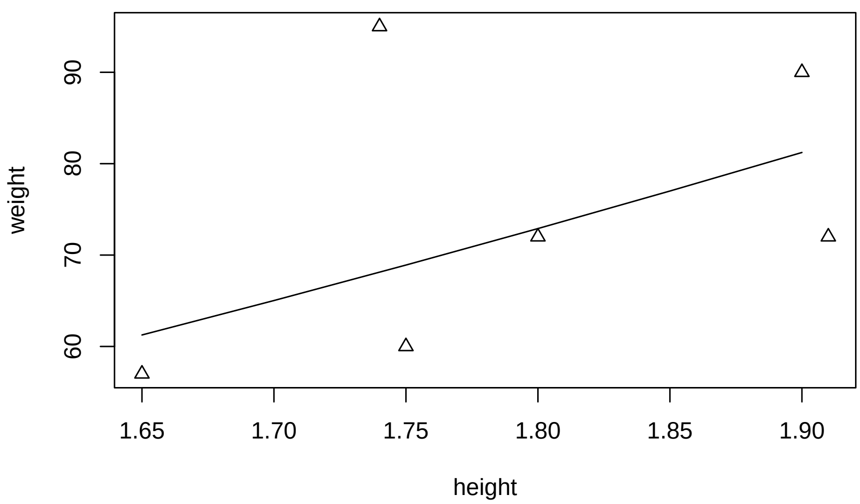
2.15 Plot an expected Line for weights at BMI of 22.5

- The term for weight at BMI of 22.5 is:

```
22.5*(height)^2
```

- This term can be used to compute the line of expected weights at a BMI of 22.5 using the lines() function
- As a note, we can't use the lines() function without first creating a plot in R.

```
plot(height, weight, pch =2)
hh <- c(1.65, 1.70, 1.75, 1.80, 1.85, 1.90)
lines(hh, 22.5 * (hh)^2)
```



2.16 Vectors

- The weight and height vectors are called numeric vectors
- Besides numeric vectors, there are numeric and character vectors

2.17 Using other libraries

2.17.1 The iris data-frame

- Now we have a better understanding of what R can give us, let us use another library more commonly used datasets
- At times, it is possible that you will need to figure out different ways to install a library to use it

```r
library(datasets)
```

- In the next chunk, we access the iris data, and look at a summary of the dataset

```r
data(iris)
summary(iris)
```

```
##   Sepal.Length    Sepal.Width     Petal.Length    Petal.Width   
##  Min.   :4.300   Min.   :2.000   Min.   :1.000   Min.   :0.100  
##  1st Qu.:5.100   1st Qu.:2.800   1st Qu.:1.600   1st Qu.:0.300  
##  Median :5.800   Median :3.000   Median :4.350   Median :1.300  
##  Mean   :5.843   Mean   :3.057   Mean   :3.758   Mean   :1.199  
##  3rd Qu.:6.400   3rd Qu.:3.300   3rd Qu.:5.100   3rd Qu.:1.800  
##  Max.   :7.900   Max.   :4.400   Max.   :6.900   Max.   :2.500  
##        Species  
##  setosa    :50  
##  versicolor:50  
##  virginica :50  
##                 
##                 
## 
```

- Another form to look at the iris data-frame is typing its name

```r
iris
```

```
##     Sepal.Length Sepal.Width Petal.Length Petal.Width    Species
## 1            5.1         3.5          1.4         0.2     setosa
## 2            4.9         3.0          1.4         0.2     setosa
## 3            4.7         3.2          1.3         0.2     setosa
## 4            4.6         3.1          1.5         0.2     setosa
## 5            5.0         3.6          1.4         0.2     setosa
## 6            5.4         3.9          1.7         0.4     setosa
## 7            4.6         3.4          1.4         0.3     setosa
## 8            5.0         3.4          1.5         0.2     setosa
## 9            4.4         2.9          1.4         0.2     setosa
## 10           4.9         3.1          1.5         0.1     setosa
## 11           5.4         3.7          1.5         0.2     setosa
## 12           4.8         3.4          1.6         0.2     setosa
## 13           4.8         3.0          1.4         0.1     setosa
## 14           4.3         3.0          1.1         0.1     setosa
## 15           5.8         4.0          1.2         0.2     setosa
```

```
## 16          5.7         4.4          1.5         0.4     setosa
## 17          5.4         3.9          1.3         0.4     setosa
## 18          5.1         3.5          1.4         0.3     setosa
## 19          5.7         3.8          1.7         0.3     setosa
## 20          5.1         3.8          1.5         0.3     setosa
## 21          5.4         3.4          1.7         0.2     setosa
## 22          5.1         3.7          1.5         0.4     setosa
## 23          4.6         3.6          1.0         0.2     setosa
## 24          5.1         3.3          1.7         0.5     setosa
## 25          4.8         3.4          1.9         0.2     setosa
## 26          5.0         3.0          1.6         0.2     setosa
## 27          5.0         3.4          1.6         0.4     setosa
## 28          5.2         3.5          1.5         0.2     setosa
## 29          5.2         3.4          1.4         0.2     setosa
## 30          4.7         3.2          1.6         0.2     setosa
## 31          4.8         3.1          1.6         0.2     setosa
## 32          5.4         3.4          1.5         0.4     setosa
## 33          5.2         4.1          1.5         0.1     setosa
## 34          5.5         4.2          1.4         0.2     setosa
## 35          4.9         3.1          1.5         0.2     setosa
## 36          5.0         3.2          1.2         0.2     setosa
## 37          5.5         3.5          1.3         0.2     setosa
## 38          4.9         3.6          1.4         0.1     setosa
## 39          4.4         3.0          1.3         0.2     setosa
## 40          5.1         3.4          1.5         0.2     setosa
## 41          5.0         3.5          1.3         0.3     setosa
## 42          4.5         2.3          1.3         0.3     setosa
## 43          4.4         3.2          1.3         0.2     setosa
## 44          5.0         3.5          1.6         0.6     setosa
## 45          5.1         3.8          1.9         0.4     setosa
## 46          4.8         3.0          1.4         0.3     setosa
## 47          5.1         3.8          1.6         0.2     setosa
## 48          4.6         3.2          1.4         0.2     setosa
## 49          5.3         3.7          1.5         0.2     setosa
## 50          5.0         3.3          1.4         0.2     setosa
## 51          7.0         3.2          4.7         1.4 versicolor
## 52          6.4         3.2          4.5         1.5 versicolor
## 53          6.9         3.1          4.9         1.5 versicolor
## 54          5.5         2.3          4.0         1.3 versicolor
## 55          6.5         2.8          4.6         1.5 versicolor
## 56          5.7         2.8          4.5         1.3 versicolor
## 57          6.3         3.3          4.7         1.6 versicolor
## 58          4.9         2.4          3.3         1.0 versicolor
## 59          6.6         2.9          4.6         1.3 versicolor
## 60          5.2         2.7          3.9         1.4 versicolor
## 61          5.0         2.0          3.5         1.0 versicolor
```

```
## 62          5.9         3.0          4.2         1.5 versicolor
## 63          6.0         2.2          4.0         1.0 versicolor
## 64          6.1         2.9          4.7         1.4 versicolor
## 65          5.6         2.9          3.6         1.3 versicolor
## 66          6.7         3.1          4.4         1.4 versicolor
## 67          5.6         3.0          4.5         1.5 versicolor
## 68          5.8         2.7          4.1         1.0 versicolor
## 69          6.2         2.2          4.5         1.5 versicolor
## 70          5.6         2.5          3.9         1.1 versicolor
## 71          5.9         3.2          4.8         1.8 versicolor
## 72          6.1         2.8          4.0         1.3 versicolor
## 73          6.3         2.5          4.9         1.5 versicolor
## 74          6.1         2.8          4.7         1.2 versicolor
## 75          6.4         2.9          4.3         1.3 versicolor
## 76          6.6         3.0          4.4         1.4 versicolor
## 77          6.8         2.8          4.8         1.4 versicolor
## 78          6.7         3.0          5.0         1.7 versicolor
## 79          6.0         2.9          4.5         1.5 versicolor
## 80          5.7         2.6          3.5         1.0 versicolor
## 81          5.5         2.4          3.8         1.1 versicolor
## 82          5.5         2.4          3.7         1.0 versicolor
## 83          5.8         2.7          3.9         1.2 versicolor
## 84          6.0         2.7          5.1         1.6 versicolor
## 85          5.4         3.0          4.5         1.5 versicolor
## 86          6.0         3.4          4.5         1.6 versicolor
## 87          6.7         3.1          4.7         1.5 versicolor
## 88          6.3         2.3          4.4         1.3 versicolor
## 89          5.6         3.0          4.1         1.3 versicolor
## 90          5.5         2.5          4.0         1.3 versicolor
## 91          5.5         2.6          4.4         1.2 versicolor
## 92          6.1         3.0          4.6         1.4 versicolor
## 93          5.8         2.6          4.0         1.2 versicolor
## 94          5.0         2.3          3.3         1.0 versicolor
## 95          5.6         2.7          4.2         1.3 versicolor
## 96          5.7         3.0          4.2         1.2 versicolor
## 97          5.7         2.9          4.2         1.3 versicolor
## 98          6.2         2.9          4.3         1.3 versicolor
## 99          5.1         2.5          3.0         1.1 versicolor
## 100         5.7         2.8          4.1         1.3 versicolor
## 101         6.3         3.3          6.0         2.5  virginica
## 102         5.8         2.7          5.1         1.9  virginica
## 103         7.1         3.0          5.9         2.1  virginica
## 104         6.3         2.9          5.6         1.8  virginica
## 105         6.5         3.0          5.8         2.2  virginica
## 106         7.6         3.0          6.6         2.1  virginica
## 107         4.9         2.5          4.5         1.7  virginica
```

```
## 108          7.3         2.9          6.3         1.8  virginica
## 109          6.7         2.5          5.8         1.8  virginica
## 110          7.2         3.6          6.1         2.5  virginica
## 111          6.5         3.2          5.1         2.0  virginica
## 112          6.4         2.7          5.3         1.9  virginica
## 113          6.8         3.0          5.5         2.1  virginica
## 114          5.7         2.5          5.0         2.0  virginica
## 115          5.8         2.8          5.1         2.4  virginica
## 116          6.4         3.2          5.3         2.3  virginica
## 117          6.5         3.0          5.5         1.8  virginica
## 118          7.7         3.8          6.7         2.2  virginica
## 119          7.7         2.6          6.9         2.3  virginica
## 120          6.0         2.2          5.0         1.5  virginica
## 121          6.9         3.2          5.7         2.3  virginica
## 122          5.6         2.8          4.9         2.0  virginica
## 123          7.7         2.8          6.7         2.0  virginica
## 124          6.3         2.7          4.9         1.8  virginica
## 125          6.7         3.3          5.7         2.1  virginica
## 126          7.2         3.2          6.0         1.8  virginica
## 127          6.2         2.8          4.8         1.8  virginica
## 128          6.1         3.0          4.9         1.8  virginica
## 129          6.4         2.8          5.6         2.1  virginica
## 130          7.2         3.0          5.8         1.6  virginica
## 131          7.4         2.8          6.1         1.9  virginica
## 132          7.9         3.8          6.4         2.0  virginica
## 133          6.4         2.8          5.6         2.2  virginica
## 134          6.3         2.8          5.1         1.5  virginica
## 135          6.1         2.6          5.6         1.4  virginica
## 136          7.7         3.0          6.1         2.3  virginica
## 137          6.3         3.4          5.6         2.4  virginica
## 138          6.4         3.1          5.5         1.8  virginica
## 139          6.0         3.0          4.8         1.8  virginica
## 140          6.9         3.1          5.4         2.1  virginica
## 141          6.7         3.1          5.6         2.4  virginica
## 142          6.9         3.1          5.1         2.3  virginica
## 143          5.8         2.7          5.1         1.9  virginica
## 144          6.8         3.2          5.9         2.3  virginica
## 145          6.7         3.3          5.7         2.5  virginica
## 146          6.7         3.0          5.2         2.3  virginica
## 147          6.3         2.5          5.0         1.9  virginica
## 148          6.5         3.0          5.2         2.0  virginica
## 149          6.2         3.4          5.4         2.3  virginica
## 150          5.9         3.0          5.1         1.8  virginica
```

2.17.2 More Visualization of the iris dataset

These are basic R functions that allow us to get general information about a dataframe:

- dim()
- head()
- View()
- class()
- str()

2.17.3 Basic R functions

```
dim(iris)
head(iris)
View(iris)
class(iris)
str(iris)
```

- Select the number of lines to visualize with the head command

```
head(iris, n = 10)
```

```
##    Sepal.Length Sepal.Width Petal.Length Petal.Width Species
## 1           5.1         3.5          1.4         0.2  setosa
## 2           4.9         3.0          1.4         0.2  setosa
## 3           4.7         3.2          1.3         0.2  setosa
## 4           4.6         3.1          1.5         0.2  setosa
## 5           5.0         3.6          1.4         0.2  setosa
## 6           5.4         3.9          1.7         0.4  setosa
## 7           4.6         3.4          1.4         0.3  setosa
## 8           5.0         3.4          1.5         0.2  setosa
## 9           4.4         2.9          1.4         0.2  setosa
## 10          4.9         3.1          1.5         0.1  setosa
```

- Determine the number of columns and rows in the dataframe:

```
dim(iris)
```

```
## [1] 150   5
```

- Determine the class() that the dataset belongs to

```
class(iris)
```

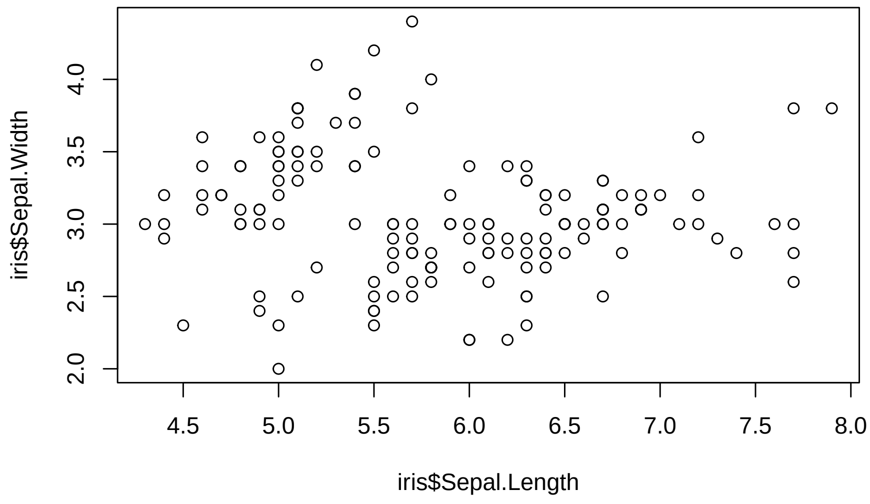
```
## [1] "data.frame"
```

2.18 More Visualization

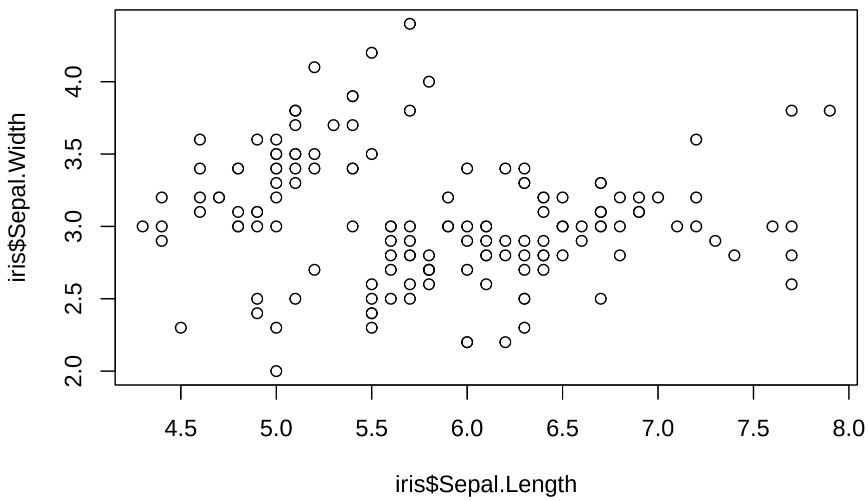
2.18.1 Scatterplot (includes warnings)

```
plot(data=iris, iris$Sepal.Length, iris$Sepal.Width) ## R will complain about this com
```

```
## Warning in plot.window(...): "data" is not a graphical parameter

## Warning in plot.xy(xy, type, ...): "data" is not a graphical parameter

## Warning in axis(side = side, at = at, labels = labels, ...): "data" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "data" is not a
## graphical parameter

## Warning in box(...): "data" is not a graphical parameter

## Warning in title(...): "data" is not a graphical parameter
```



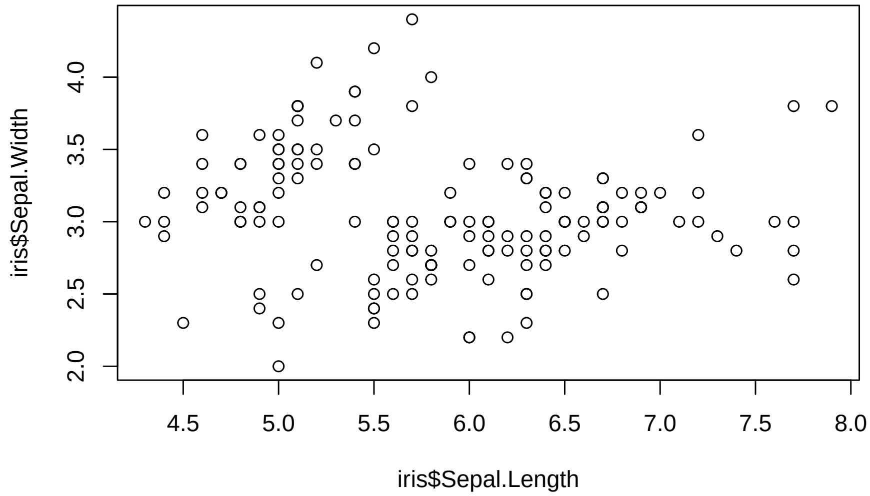
```
plot(iris$Sepal.Length, iris$Sepal.Width) ## According to the error, this was plotted using a di
```



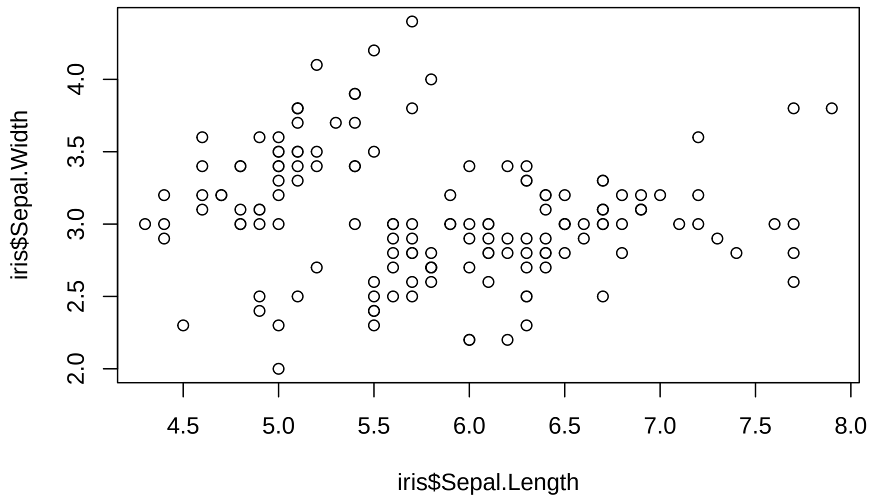
2.18.2 Scatterplot

Plot without warning and message

```
plot(data=iris, iris$Sepal.Length, iris$Sepal.Width) ## R will complain about this command
```

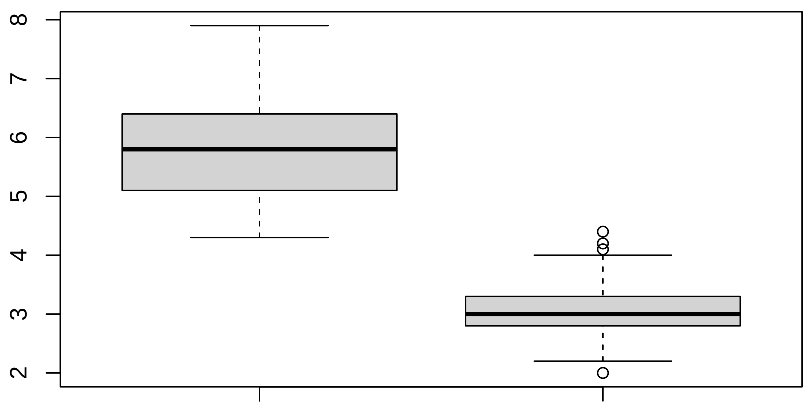



```
plot(iris$Sepal.Length, iris$Sepal.Width) ## According to the error, this was plotted t
```



2.18.3 Boxplot

```
boxplot(data=iris, iris$Sepal.Length, iris$Sepal.Width)
```

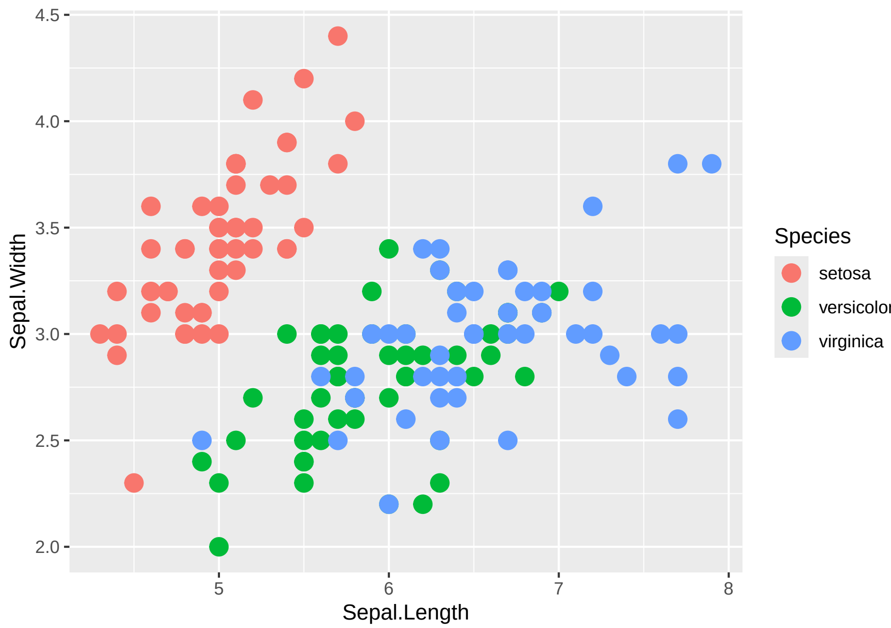


2.19 Data Visualization with specialized libraries

- In R, there are packages designed with the purpose of making good-looking graphics. This is the case with the ggplot2. The chunk below installs the ggplot2 library, loads the library into the R environment and then plots the data present in the iris data-frame.
- You can uncomment the installation line if you need to install it

2.19.1 Ggplot2

```
#install.packages("ggplot2")
library(ggplot2)
ggplot(data=iris, aes(x=Sepal.Length, y=Sepal.Width, color=Species)) + geom_point(size=4)
```



In our activities to visualize human genomic data, we will use a library called qqman, to visualize the biological association, through a plot known as the Manhattan plot.

2.19.2 Visualizing GWAS

- The simplest definition of a GWAS is the statistical or significant association between a phenotype (trait) and a genotype. This association can also be called *biological association*.
- Information about association of SNPs with Huntington's Disease can be found at the Chaves 2019 Huntington's disease paper

2.19.2.1 Package installation

- We need to install and load package qqman

```r
## install.packages("qqman")
library(qqman)
```

```
##
```

```
## For example usage please run: vignette('qqman')
```

```
##
## Citation appreciated but not required:
## Turner, (2018). qqman: an R package for visualizing GWAS results using Q-Q and manhattan plots
##
```

2.19.2.2 Assign text file to dataframe object

- After, we load data-frame to be visualized
- Exact location of text file in your system needs to be determined

```
GWAS_TABLE <- read.table("./data/chr4.txt",
                         sep = " ",
                         header = T)
```

2.19.2.3 Get information about object with head()

- Use head() function to inspect the data-frame just loaded

```
head(GWAS_TABLE, n = 10)
```

```
##    CHR         SNP     BP         P
## 1    4 chr4:128096 128096 0.0133500
## 2    4 chr4:516586 516586 0.0076260
## 3    4 chr4:523979 523979 0.0024960
## 4    4 chr4:527217 527217 0.0217400
## 5    4 chr4:566177 566177 0.0008988
## 6    4 chr4:578679 578679 0.0162100
## 7    4 chr4:578790 578790 0.0103700
## 8    4 chr4:579307 579307 0.0334600
## 9    4 chr4:580259 580259 0.0190400
## 10   4 chr4:585318 585318 0.0317600
```

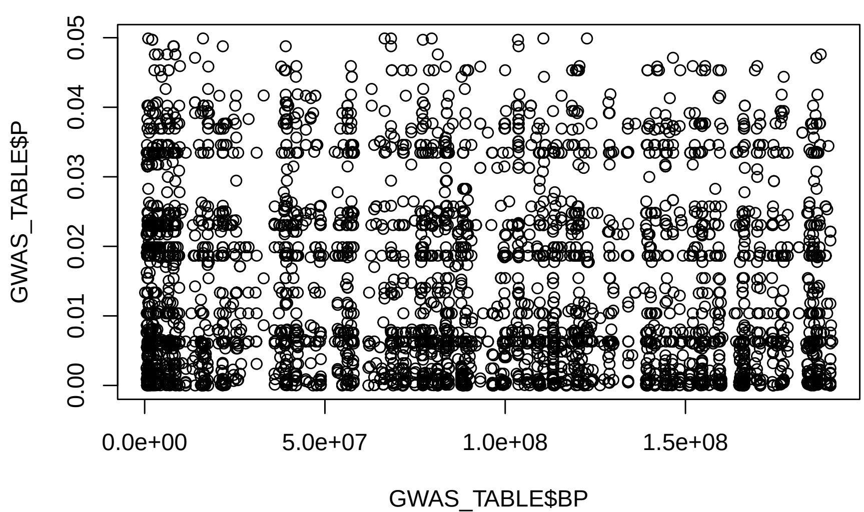
2.19.2.4 Plotting GWAS data-frame

- In this section we use functions plot() and boxplot() to visualize data-frame
- In the y access, we see genomic coordinates and in the x access, p-valoues of the biological association

- Note that depending on the synthax used for plotting, R may complain
- Comments not turned off

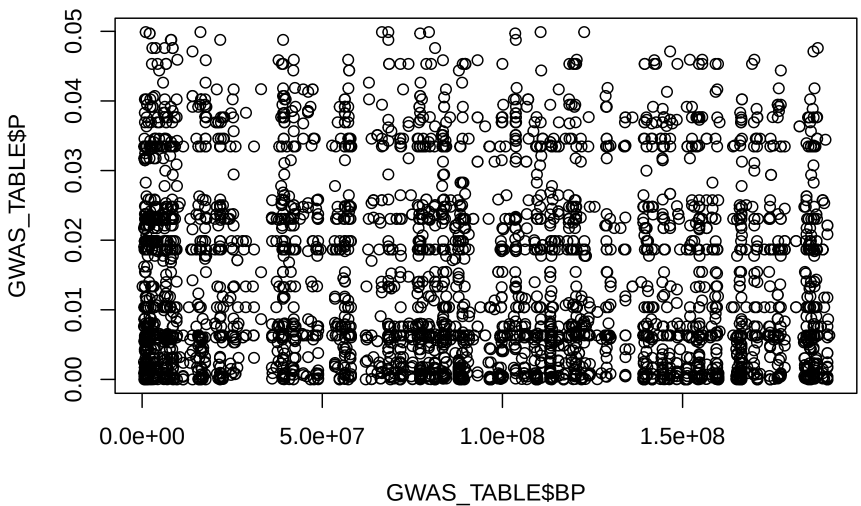
```
plot(data=GWAS_TABLE, GWAS_TABLE$BP, GWAS_TABLE$P) ## R complains
```

```
## Warning in plot.window(...): "data" is not a graphical parameter

## Warning in plot.xy(xy, type, ...): "data" is not a graphical parameter

## Warning in axis(side = side, at = at, labels = labels, ...): "data" is not a
## graphical parameter
## Warning in axis(side = side, at = at, labels = labels, ...): "data" is not a
## graphical parameter

## Warning in box(...): "data" is not a graphical parameter

## Warning in title(...): "data" is not a graphical parameter
```



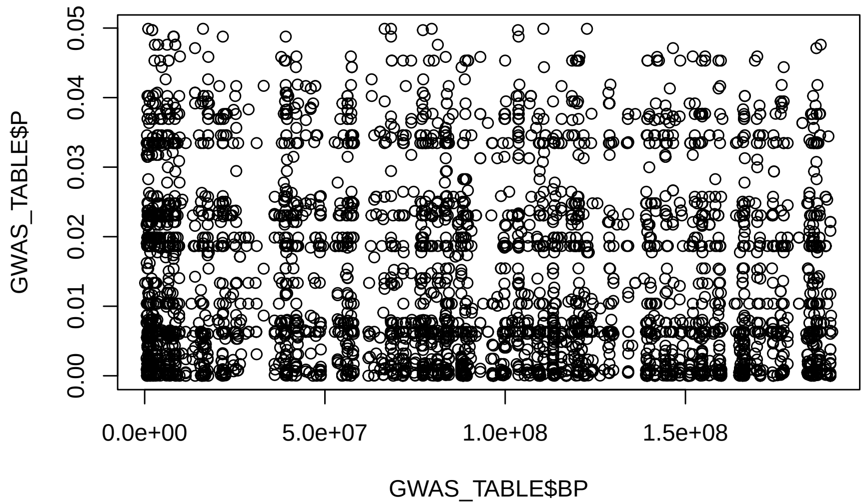
- Turn off comments

```
plot(data=GWAS_TABLE, GWAS_TABLE$BP, GWAS_TABLE$P) ## R complains
```



2.19.2.5 Plotting GWAS data-frame

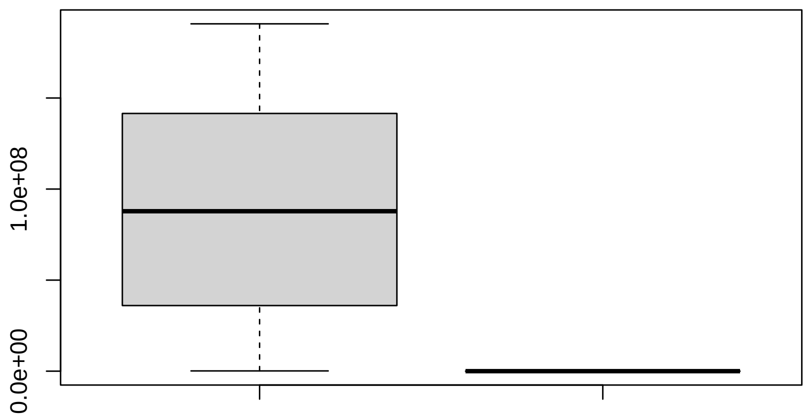
```
plot(GWAS_TABLE$BP, GWAS_TABLE$P) ## R does not complain
```



2.19.2.6 Boxplot

- Boxplot

```
boxplot(data=GWAS_TABLE, GWAS_TABLE$BP, GWAS_TABLE$P)
```



- Compare the chromosomal coordinates and the values of the p-values in the boxplot above
- The chunk below allows one to ommit NA values in data-frame

```
GWAS_TABLE_Ommit <- na.omit(GWAS_TABLE)
```

A seguir, criamos uma variável que armazena as posições das SNPs a serem realçadas em verde no Manhattan plot. Estas SNPs são mutações biológica e estatisticamente associadas à Doença de Huntington no gene de uma sortilina, localizada em proximidade física ao gene da proteína huntingtina mutada, a qual é a causadora medeliana (segue as leis de segregação genética de Mendel) da Doença de Huntington.

2.20 Create vector to highlight genome coordinates in Manhattan plot

```
SNP_HIGHLIGHT <- c("chr4:3043512","chr4:3043513","chr4:3048207","chr4:3224216",
                   "chr4:3231772","chr4:3233844","chr4:3235081","chr4:3235084",
                   "chr4:3236881","chr4:3236883","chr4:3241845","chr4:3243804",
                   "chr4:3263138","chr4:3265130","chr4:3265710","chr4:3314646",
```

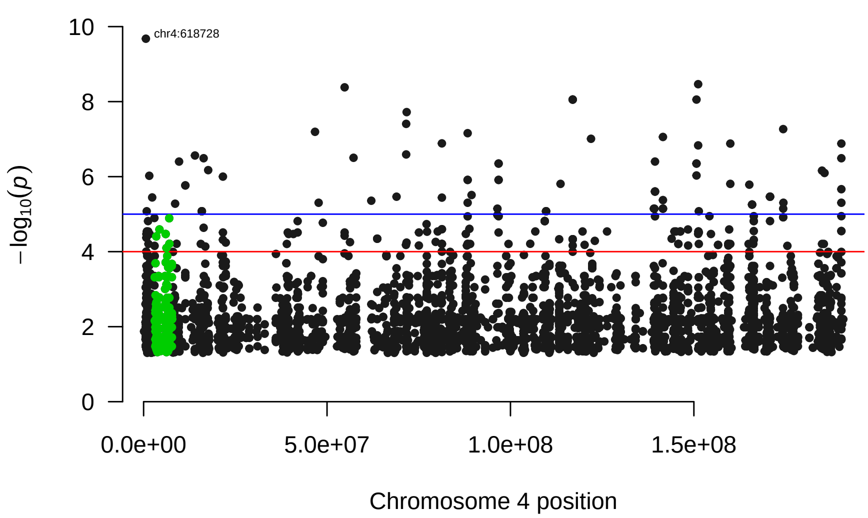
```
"chr4:3380088","chr4:3409359","chr4:3411110","chr4:3415336",
"chr4:3415378","chr4:3438643","chr4:3446091","chr4:3449886",
"chr4:3473066","chr4:3476809","chr4:3480439","chr4:3487151",
"chr4:3496058","chr4:3496110","chr4:3506933","chr4:3508752",
"chr4:3510957","chr4:3512690","chr4:3517746","chr4:3518190",
"chr4:3529671","chr4:3532327","chr4:3533066","chr4:3746133",
"chr4:3747842","chr4:3748134","chr4:3765305","chr4:3765336",
"chr4:3944253","chr4:3944752","chr4:3944888","chr4:3946166",
"chr4:3946175","chr4:3969218","chr4:4051294","chr4:4076788",
"chr4:4103104","chr4:4103105","chr4:4109198","chr4:4109210",
"chr4:4240627","chr4:4242705","chr4:4243668","chr4:4245210",
"chr4:4245510","chr4:4245513","chr4:4245591","chr4:4245926",
"chr4:4245929","chr4:4246109","chr4:4246433","chr4:4246453",
"chr4:4246457","chr4:4246497","chr4:4249414","chr4:4249415",
"chr4:4249484","chr4:4271623","chr4:4275306","chr4:4304749",
"chr4:4318931","chr4:4318970","chr4:4319564","chr4:4319728",
"chr4:4319750","chr4:4322078","chr4:4709657","chr4:4732282",
"chr4:4789635","chr4:4822960","chr4:4824890","chr4:4825092",
"chr4:4825180","chr4:4865316","chr4:4865321","chr4:5018702",
"chr4:5812778","chr4:5814082","chr4:5833660","chr4:5833899",
"chr4:5835541","chr4:5851205","chr4:5862752","chr4:5862938",
"chr4:5862943","chr4:5901873","chr4:5905499","chr4:5906287",
"chr4:6018891","chr4:6019046","chr4:6020190","chr4:6020367",
"chr4:6025638","chr4:6025656","chr4:6025766","chr4:6026058",
"chr4:6083488","chr4:6204935","chr4:6235553","chr4:6237142",
"chr4:6238466","chr4:6239906","chr4:6240929","chr4:6245022",
"chr4:6245618","chr4:6245732","chr4:6245915","chr4:6246075",
"chr4:6246373","chr4:6246959","chr4:6290594","chr4:6292020",
"chr4:6294095","chr4:6298375","chr4:6316092","chr4:6321396",
"chr4:6324647","chr4:6324785","chr4:6327669","chr4:6328354",
"chr4:6328507","chr4:6333130","chr4:6333559","chr4:6333669",
"chr4:6335966","chr4:6435341","chr4:6435486","chr4:6435926",
"chr4:6437191","chr4:6437197","chr4:6457121","chr4:6457131",
"chr4:6457132","chr4:6568390","chr4:6570032","chr4:6570768",
"chr4:6596360","chr4:6613252","chr4:6613462","chr4:6620991",
"chr4:6624771","chr4:6626154","chr4:6641969","chr4:6642090",
"chr4:6644466","chr4:6644467","chr4:6644468","chr4:6647889",
"chr4:6648300","chr4:6662665","chr4:6663319","chr4:6663715",
"chr4:6674554","chr4:6678553","chr4:6678599","chr4:6690535",
"chr4:6698664","chr4:6698667","chr4:6698706","chr4:6720572",
"chr4:6911679","chr4:6985889","chr4:6987394","chr4:7002344",
"chr4:7004495","chr4:7004506","chr4:7005196","chr4:7005199",
"chr4:7024077","chr4:7024398","chr4:7029430","chr4:7031064",
"chr4:7044357","chr4:7044380","chr4:7048842","chr4:7052115",
"chr4:7055253","chr4:7064243","chr4:7067765","chr4:7073187",
```

```
                    "chr4:7074027","chr4:7677967","chr4:7701947","chr4:7702795",
                    "chr4:7703505","chr4:7703807","chr4:7704795","chr4:7704818",
                    "chr4:7709703","chr4:7712150","chr4:7714490","chr4:7733843",
                    "chr4:7735162","chr4:7735164","chr4:7736103","chr4:7736112")
```

2.21 Biological Association Visualization: Manhattan Plot

- Finally, we plot the Manhattanh plot graph

```
manhattan(GWAS_TABLE_Ommit,
          highlight = SNP_HIGHLIGHT,
          annotateTop = T,
          annotatePval = 0.20,
          genomewideline = -log10(0.0001))
```



2.22 References

2.23 Session Info

```r
sessionInfo()
```

```
## R version 4.4.1 (2024-06-14)
## Platform: aarch64-apple-darwin20
## Running under: macOS 15.7.4
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/Chicago
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] qqman_0.1.9   ggplot2_3.5.1
##
## loaded via a namespace (and not attached):
##  [1] vctrs_0.6.5       cli_3.6.3         knitr_1.48        rlang_1.1.4
##  [5] xfun_0.48         highr_0.11        generics_0.1.3    calibrate_1.7.7
##  [9] labeling_0.4.3    glue_1.8.0        colorspace_2.1-1  htmltools_0.5.8.1
## [13] tinytex_0.53      scales_1.3.0      fansi_1.0.6       rmarkdown_2.28
## [17] grid_4.4.1        evaluate_1.0.0    munsell_0.5.1     tibble_3.2.1
## [21] MASS_7.3-61       fastmap_1.2.0     yaml_2.3.10       lifecycle_1.0.4
## [25] bookdown_0.40     compiler_4.4.1    dplyr_1.1.4       pkgconfig_2.0.3
## [29] rstudioapi_0.16.0 farver_2.1.2      digest_0.6.37     R6_2.5.1
## [33] tidyselect_1.2.1  utf8_1.2.4        pillar_1.9.0      magrittr_2.0.3
## [37] withr_3.0.1       tools_4.4.1       gtable_0.3.5
```

Chapter 3

Processamento de dados públicos de neuroblastoma

3.1 Origem dos dados

Os dados analisados neste módulo têm origem na base de dados genômicos R2.

A plataforma R2 é uma ferramenta de análise e visualização genômica online de livre acesso que pode analisar uma grande coleção de dados públicos, mas também permite a análise protegida de conjuntos de dados privados, conforme indicado por Rijswijk (2024).

A R2 consiste em um banco de dados que armazena as informações genômicas, acoplado a um amplo conjunto de ferramentas para analisar/visualizar os conjuntos de dados.

Mais informações sobre a plataforma R2 pode ser encontradas no site da plataforma:

https://hgserver1.amc.nl/cgi-bin/r2/main.cgi

3.2 Carregando dados pre-processados de R2

Nesta parte, vamos carregar uma dataframe contendo dados baixados de R2 em que já fizemos pre-processamento. Vamos gerar algumas visualizações para nos familiarizarmos com os dados. O objeto r2_gse62564_GSVA_Metadata.rds deve ser baixado da pasta no Google Drive no seguinte link: https://drive.google.com/drive/folders/1Kkos4HK2EP4ntrQOJE0mrHYlpeMa1yi4?usp=sharing

Uma vez que o arquivo é baixado, pode-se carregá-lo no ambiente R usando a função readRDS, como feito abaixo:

```
r2_gse62564_GSVA_Metadata <- readRDS("./data/r2_gse62564_GSVA_Metadata.rds")
```

No neuroblastoma, o principal sistema de classificação de risco classifica os pacientes usando características como idade, estadiamento, histologia e status de amplificação do MYCN.

O prognóstico é a associação entre as características do tumor e a sobrevida do paciente.

O MYCN é um gene que codifica um fator de transcrição muito importante no prognóstico do tumor.

Vamos analisar a expressão do MYCN comparada pelo grupo de risco do neuroblastoma.

Primeiro, vamos ver que tipo de variável é o status do MYCN e a expressão do MYCN.

Verificar a classe da variável mycn_status:

```
class(r2_gse62564_GSVA_Metadata$mycn_status)
```

```
## [1] "character"
```

Verificar a classe da variável MYCN:

```
class(r2_gse62564_GSVA_Metadata$MYCN)
```

```
## [1] "character"
```

Ambos são do tipo "character". Para plotar, precisamos fazer com que a expressão MYCN seja do tipo numérico:

```
r2_gse62564_GSVA_Metadata$MYCN <- as.numeric(
  r2_gse62564_GSVA_Metadata$MYCN)
```

Verificar novamente a classe da variável MYCN:

```
class(r2_gse62564_GSVA_Metadata$MYCN)
```

```
## [1] "numeric"
```

Agora a classe de MYCN é um vetor numérico.

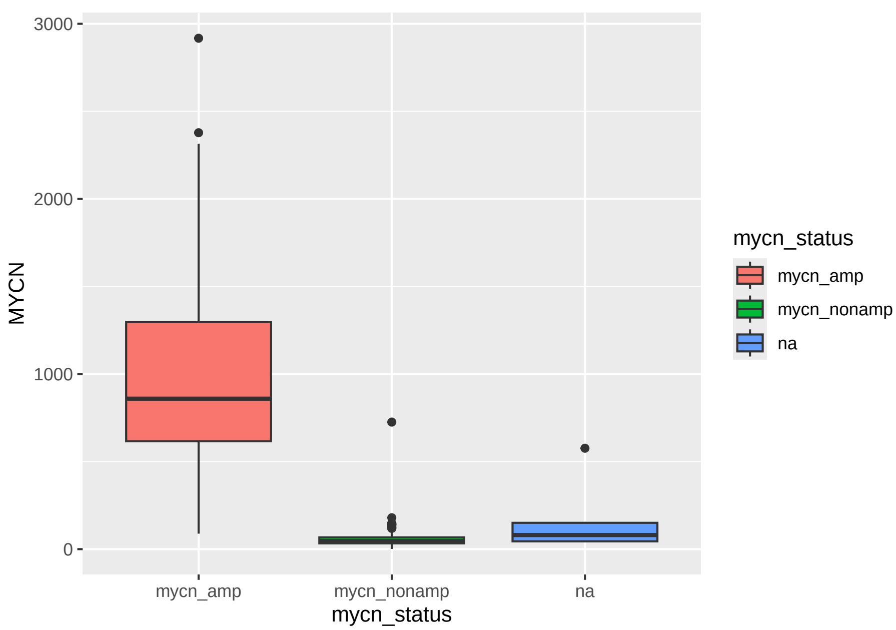
3.3 Primeiros gráficos com dados de neuroblastoma

3.3.1 Mycn_status vs Expressão de MYCN

Vamos plotar uma figura com a variável mycn_status no eixo X e expressão de MYCN no eixo Y.

Vamos tentar ver se o ggplot, que usamos antes, considera a variável mycn_status categórica como ela é agora.

```
library(ggplot2)
ggplot(r2_gse62564_GSVA_Metadata, aes(x=mycn_status, y=MYCN,
                                     fill=mycn_status)) +
  geom_boxplot()
```

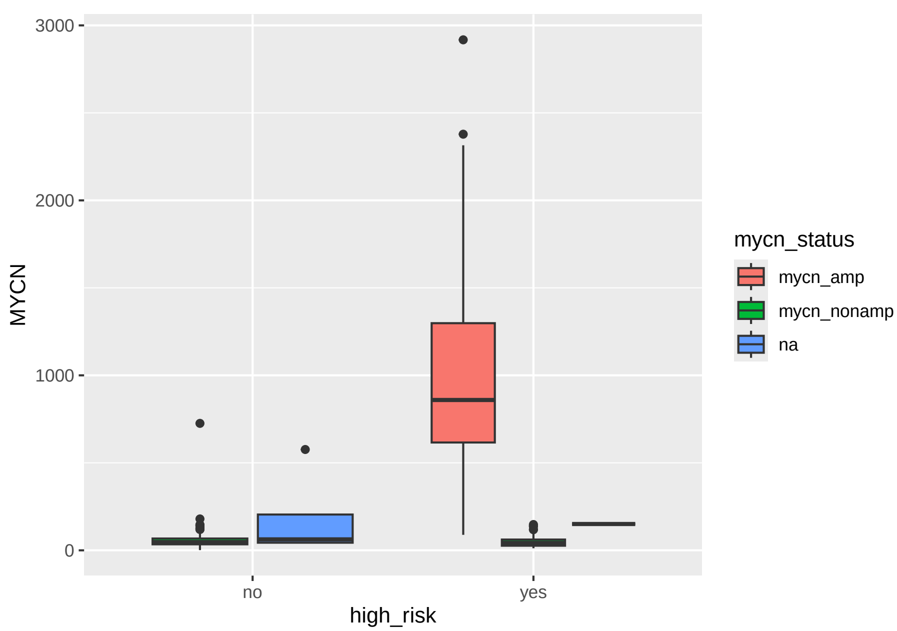


Vemos que existem três categorias de mycn_status: mycn_amp, mycn_nonamp e nas. mycn_amp significa amplificação de MYCN, um fator de mau prognóstico. mycn_nonamp significa ausência de amplificação de MYCN. NA significa que não há informações sobre o status de MYCN. Vemos que a amplificação de MYCN está altamente associada à expressão de MYCN, no eixo Y. Por enquanto, não vamos nos preocupar nem com a formatação dos rótulos X e Y nem com os valores N/A do status de amplificação MYCN.

3.3.2 Alto Risco vs expressão de MYCN

Podemos plotar o status de alto risco vs. amplificação MYCN e deixar o argumento de preenchimento como mycn_status

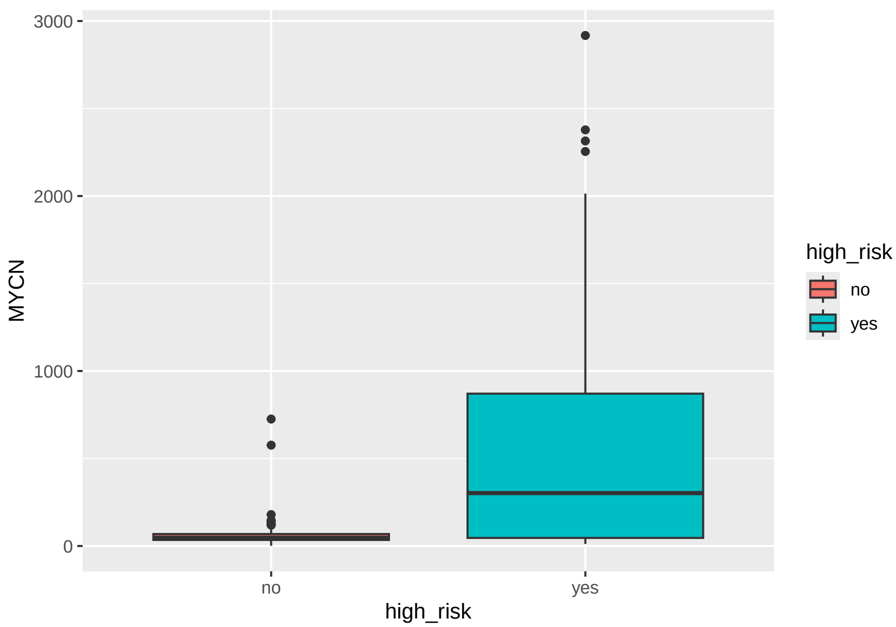
```
library(ggplot2)
ggplot(r2_gse62564_GSVA_Metadata, aes(x=high_risk, y=MYCN,
                                       fill=mycn_status)) +
    geom_boxplot()
```



Aqui vemos que a maior parte dos pacientes de alto risco (High Risk) apresentam também amplificação de MYCN e que nesses pacientes, a expressão de MYCN é alta.

Podemos ainda querer ver apenas a expressão de MYCN no grupo de alto risco, usando o parâmetro fill=high_risk:

```
library(ggplot2)
ggplot(r2_gse62564_GSVA_Metadata, aes(x=high_risk, y=MYCN,
                                      fill=high_risk)) +
    geom_boxplot()
```



Onde confirmamos que a expressão de MYCN é maior nos pacientes de alto risco (High Risk).

Chapter 4

Análise de Sobrevivência

Na análise de sobrevivência estudamos como varia a probabilidade de sobrevivência dos pacientes em função do tempo transcorrido desde o início de um estudo. Na atividade proposta para este módulo, vamos avaliar como é a probabilidade de sobrevivência de pacientes de neuroblastoma em função do status de alto risco (se um paciente é de alto risco ou baixo risco). Para dar prosseguimento, carregamos a dataframe de neuroblastoma no ambiente R, como fizemos no módulo anterior:

```
r2_gse62564_GSVA_Metadata <- readRDS( "./data/r2_gse62564_GSVA_Metadata.rds")
```

4.1 Processamento de variáveis para permitir a construção de gráficos

Precisamos definir o status do OS (Overall Survival = sobrevivência geral) como numérico, em vez de um caractere. Fazemos o mesmo para o status EFS (Event Free Survival = sobrevivência livre de evento).

Usamos o método ifelse para construir os_bin_numeric a partir de os_bin, introduzindo números ao invés da palavra evento e não evento. No caso do Overall Survival (OS), a morte do paciente é o evento.

Para usar o comando a seguir, precisamos de usar, antes, uma biblioteca do R chamada maditr:

```
library(maditr)
```

```
##
## To select rows from data: rows(mtcars, am==0)
```

```
##
## Attaching package: 'maditr'

## The following object is masked from 'package:base':
##
##     sort_by
```

Agora podemos usar o operador pipe (%>%), que toma o resultado de um comando e passa para o próximo comando:

```
r2_gse62564_GSVA_Metadata <- r2_gse62564_GSVA_Metadata %>%
  dplyr::mutate(os_bin_numeric =
                  ifelse(r2_gse62564_GSVA_Metadata$os_bin == "event", "1", "0"))
```

Agora, fazemos os_bin_numeric numérico:

```
r2_gse62564_GSVA_Metadata["os_bin_numeric"] <- as.numeric(r2_gse62564_GSVA_Metadata$os_
```

Fazemos o mesmo para EFS:

```
## EFS
r2_gse62564_GSVA_Metadata <- r2_gse62564_GSVA_Metadata %>%
  dplyr::mutate(efs_bin_numeric =
                  ifelse(r2_gse62564_GSVA_Metadata$efs_bin == "event", "1", "0"))
```

Agora, fazemos efs_bin_numeric, numérico:

```
r2_gse62564_GSVA_Metadata["efs_bin_numeric"] <- as.numeric(r2_gse62564_GSVA_Metadata$ef
```

4.2 Fazendo a análise de sobrevivência

4.2.1 Construção da curva de sobrevivência

Usamos a função survfit do pacote em R, survival, para construir o objeto fit_os_high_risk, ajustando o modelo:

```
library(survival)
fit_os_high_risk <- survfit(Surv(as.numeric(os_day), os_bin_numeric) ~ r2_gse62564_GSVA
                   data = r2_gse62564_GSVA_Metadata)
```

Observe que os_day é uma variável contendo o número de dias até o evento de interesse.

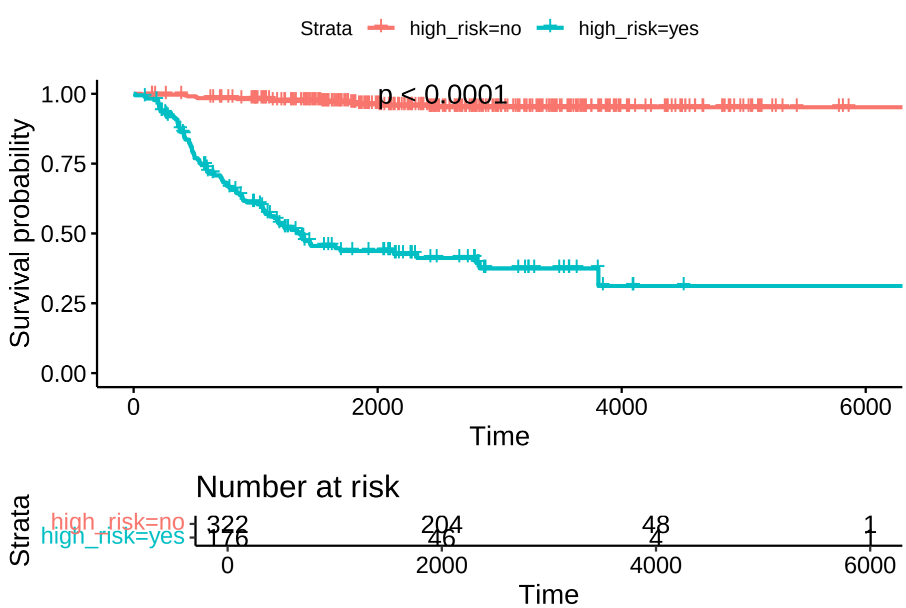
4.2.2 Plotagem da curva de sobrevivência

Usamos a função ggsurvplot do pacote em R survminer para plotar a curva de sobrevivência:

```
library(survminer)
```

```
## Loading required package: ggpubr

##
## Attaching package: 'survminer'

## The following object is masked from 'package:survival':
##
##     myeloma
```

```
ggsurvplot(fit_os_high_risk,
           data = r2_gse62564_GSVA_Metadata,
           risk.table = TRUE,
           # risk.table = "abs_pct",
           pval = TRUE,
           pval.coord = c(2000, 1),
           )
```



4.2.3 Hipóxia no microambiente tumoral

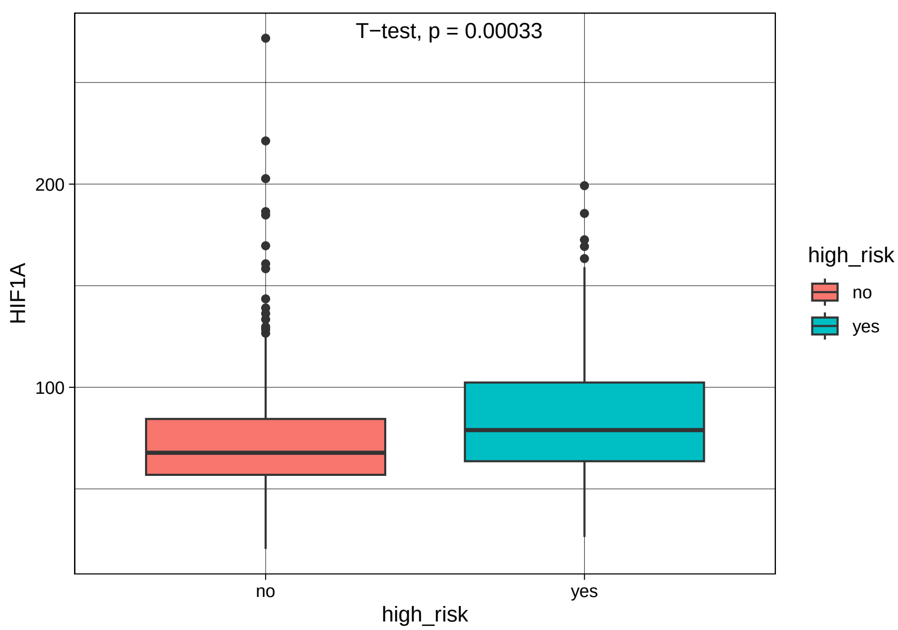
Vamos agora procurar estabelecer uma relação entre hipóxia e neuroblastoma de alto risco. Chama-se hipóxia à condição de baixa concentração de oxigênio que se estabelece no microambiente do tumor à medida que o câncer progride.

HIF1A é um fator de transcrição sensível aos níveis de oxigênio. Precisamos tornar a coluna HIF1A (variável), numérica, como fizemos com MYCN acima:

```
r2_gse62564_GSVA_Metadata$HIF1A <- as.numeric(r2_gse62564_GSVA_Metadata$HIF1A)
```

Compare as médias de HIF1A entre os grupos de alto risco usando ggplot2:

```
ggplot(r2_gse62564_GSVA_Metadata, aes(x=high_risk, y=HIF1A, fill=high_risk)) +
    geom_boxplot()+
  stat_compare_means(method = "t.test", label.x = 1.4)+
  theme_linedraw()
```



Vemos que para este conjunto de dados, a expressão de HIF1A é maior no grupo de alto risco, o que está de acordo com a noção de que aumento da hipóxia leva a maior malignidade na progressão tumoral.

4.2.4 Análise de sobrevivência do gene da hipóxia

Como vemos que indivíduos de alto risco apresentam maior expressão de HIF1A, vamos analisar também a análise de sobrevivência de HIF1A. Para isso vamos separar em dois grupos de pacientes, baseados na expressão do HIF1A. Construindo a variável HIF1A em dois quantis, um com alta expressão de HIF1A e um com baixa expressão de HIF1A. Para isso vamos usar a função ntile() do pacote tidyverse:

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v lubridate 1.9.3     v tibble    3.2.1
## v purrr     1.0.2     v tidyr     1.3.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::between()   masks maditr::between()
## x dplyr::coalesce()  masks maditr::coalesce()
## x readr::cols()      masks maditr::cols()
## x dplyr::filter()    masks stats::filter()
## x dplyr::first()     masks maditr::first()
## x dplyr::lag()       masks stats::lag()
## x dplyr::last()      masks maditr::last()
## x purrr::transpose() masks maditr::transpose()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
r2_gse62564_GSVA_Metadata <- r2_gse62564_GSVA_Metadata %>%
    mutate(quantile = ntile(HIF1A, 2))
```

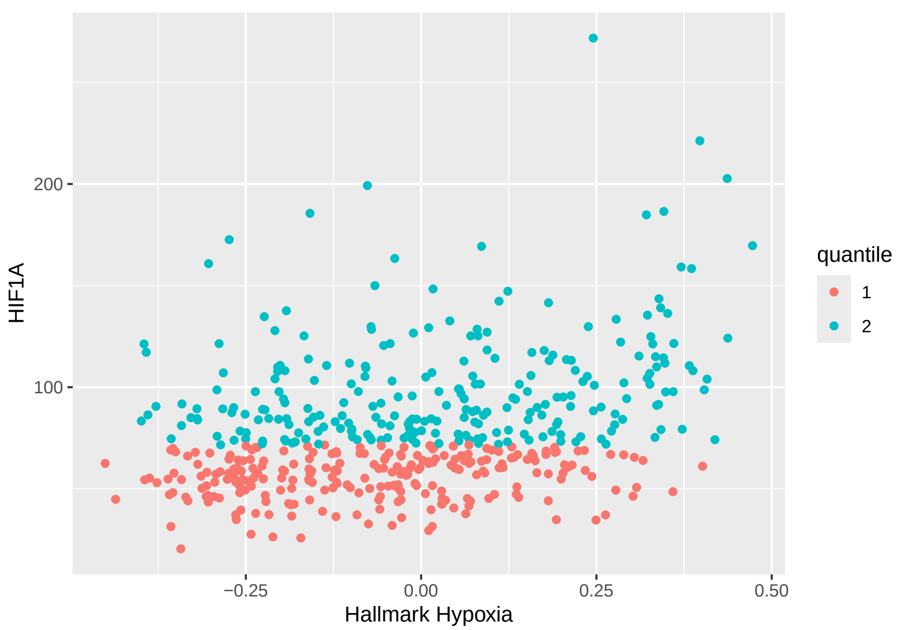
Fazendo da variável quantil HIF1A um fator

```
r2_gse62564_GSVA_Metadata$quantile <- as.factor(r2_gse62564_GSVA_Metadata$quantile )
```

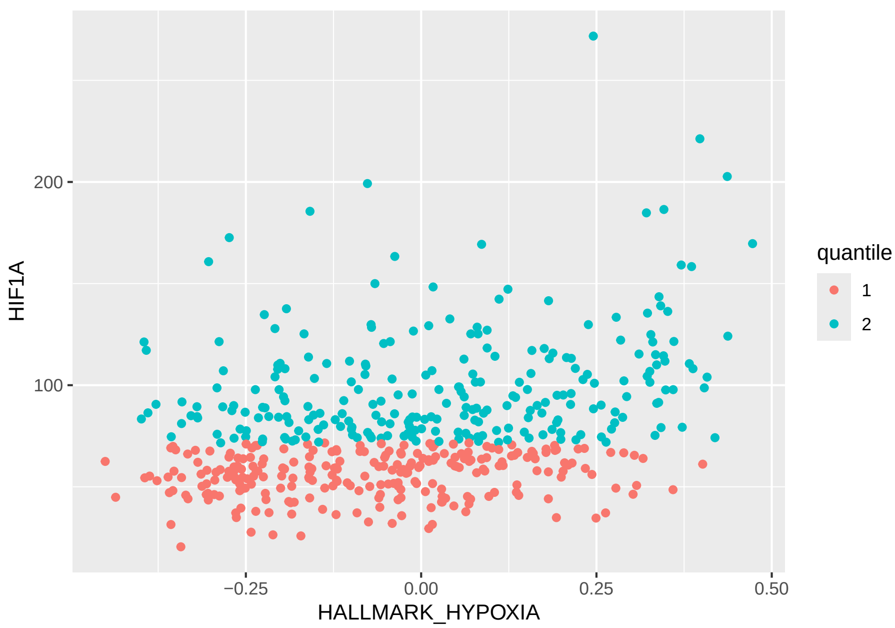
Gráfico de dispersão HALLMARK HYPOXIA vs. HIF1A

```
r2_gse62564_GSVA_Metadata$HALLMARK_HYPOXIA <- as.numeric(r2_gse62564_GSVA_Metadata$HALLMARK_HYPOX

qplot(HALLMARK_HYPOXIA, HIF1A,
      data = r2_gse62564_GSVA_Metadata,
      colour=quantile,
      ylab = "HIF1A",
      xlab = "Hallmark Hypoxia")
```

```
## Warning: `qplot()` was deprecated in ggplot2 3.4.0.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```
library(ggplot2)
# Basic scatter plot
ggplot(r2_gse62564_GSVA_Metadata, aes(x=HALLMARK_HYPOXIA, y=HIF1A, color=quantile)) + g
```



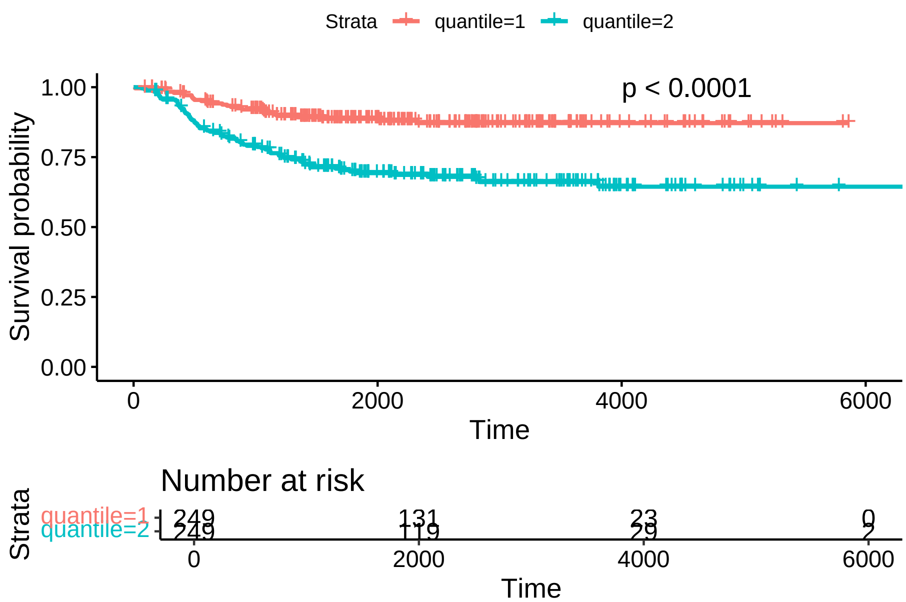
Vemos que o quantil 2 tem valores de expressão de HIF1A maiores que os valores do quantil 1.

Construímos o objeto de sobrevivência ajustado com a variável quantil HIF1A:

```
library(survival)
fit_os_HIF1A <- survfit(Surv(as.numeric(os_day), os_bin_numeric) ~ r2_gse62564_GSVA_Metadata$quar
                   data = r2_gse62564_GSVA_Metadata)
```

E finalmente traçamos a curva de sobrevivência do gráfico do Overall Survival:

```
library(survminer)
ggsurvplot(fit_os_HIF1A,
           data = r2_gse62564_GSVA_Metadata,
           risk.table = TRUE,
           # risk.table = "abs_pct",
           pval = TRUE,
           pval.coord = c(4000, 1),
           )
```



Observamos que os maiores valores na expressão de HIF1A (segundo quantil ou quantil 2. No inglês é second quantile) levam a menor probabilidade de sobrevivência. Vemos que no dataset, o segundo quantil tem menores probabilidades de sobrevivência.

4.2.4.1 Event-free survival

Construímos a variável quantil HIF1A:

```
library(tidyverse)
r2_gse62564_GSVA_Metadata <- r2_gse62564_GSVA_Metadata %>%
    mutate(quantile = ntile(HIF1A, 2))
```

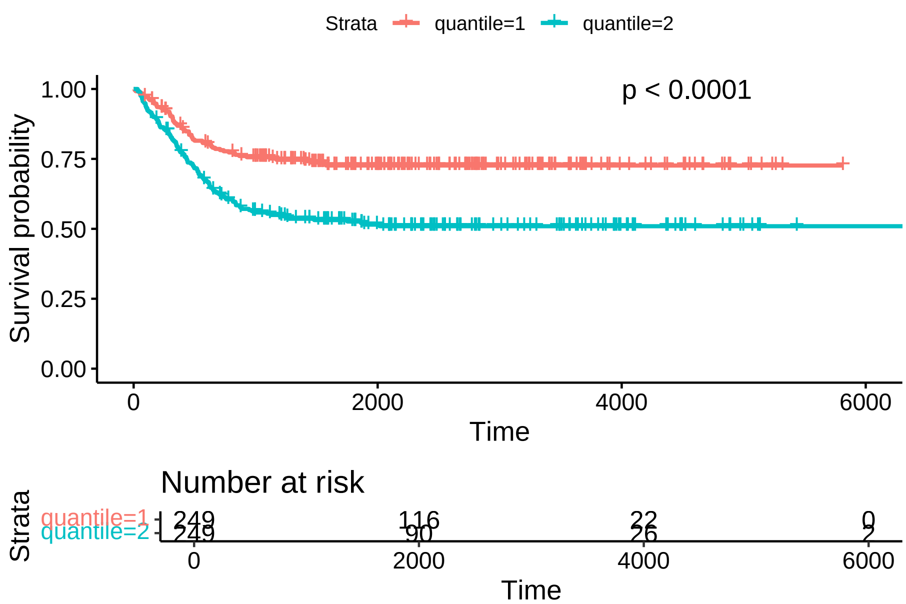
Construímos objeto de sobrevivência EFS, **fit__efs__HIF1A**

```
library(survival)
fit_efs_HIF1A <- survfit(Surv(as.numeric(efs_day), efs_bin_numeric) ~ r2_gse62564_GSVA_
                   data = r2_gse62564_GSVA_Metadata)
```

Curva de sobrevivência do gráfico EFS, usando a função **ggsurvplot**

```
library(survminer)
ggsurvplot(fit_efs_HIF1A,
```

```
           data = r2_gse62564_GSVA_Metadata,
           risk.table = TRUE,
           # risk.table = "abs_pct",
           pval = TRUE,
           pval.coord = c(4000, 1),
           )
```



4.2.5 Relação do HIF1A com a idade no diagnóstico

Precisamos criar o objeto r2_gse62564_GSVA_Metadata_Spaces_survival

A função compare_means não consegue lidar com espaços no nome da variável de exaustão de células T.

Teremos que usar um truque para lidar com os espaços em “T cell exhaustion”.

Usaremos o pacote stringr para inserir pontos (.) substituindo os espaços no dataframe r2_gse62564_GSVA_Metadata.

```
r2_gse62564_GSVA_Metadata_Spaces <- r2_gse62564_GSVA_Metadata
library(stringr)
names(r2_gse62564_GSVA_Metadata_Spaces)<-str_replace_all(names(r2_gse62564_GSVA_Metadata_Spaces),
```

Criando o objeto r2_gse62564_GSVA_Metadata_Spaces_survival:

```
r2_gse62564_GSVA_Metadata_Spaces_survival <- r2_gse62564_GSVA_Metadata_Spaces
```

Fazendo age_at_diagnosis numérico para plotar os quantis

```
r2_gse62564_GSVA_Metadata_Spaces_survival$age_at_diagnosis <- as.numeric(r2_gse62564_GS
```

Verificando a idade máxima no diagnóstico

```
max(r2_gse62564_GSVA_Metadata_Spaces_survival$age_at_diagnosis)
```

```
## [1] 8983
```

```
min(r2_gse62564_GSVA_Metadata_Spaces_survival$age_at_diagnosis)
```

```
## [1] 0
```

Construir quantis para a idade no diagnóstico (age_qtls)

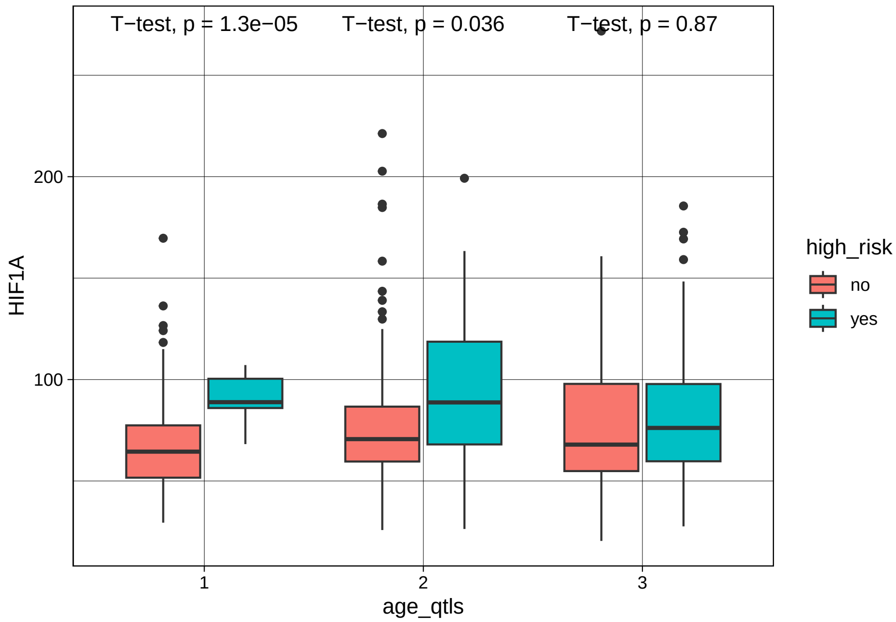
```
r2_gse62564_GSVA_Metadata_Spaces_survival_age <- r2_gse62564_GSVA_Metadata_Spaces_survi
    mutate(age_qtls = ntile(age_at_diagnosis, 3))
```

Faça age_qtls como fator

```
r2_gse62564_GSVA_Metadata_Spaces_survival_age$age_qtls <- as.factor(r2_gse62564_GSVA_Me
```

Gráfico age_qtls vs. HIF1A

```
ggplot(r2_gse62564_GSVA_Metadata_Spaces_survival_age, aes(x=age_qtls, y=HIF1A, fill=hig
    geom_boxplot()+
  stat_compare_means(method = "t.test")+
  theme_linedraw()
```


T−test, p = 1.3e−05
T−test, p = 0.036
T−test, p = 0.87
HIF1A
200
100
1
2
3
age_qtls
high_risk
no
yes

Chapter 5

Accessing the Raça/Cor Variable in DATASUS

5.1 Load libraries

```
library(microdatasus)
library(dplyr)
library(tidyverse)
library(maditr)
library(kableExtra)
```

5.2 Constrir dataframe (df) com identificação racial

```
racacor_df <- readRDS("./data/racacor_df.rds")
racacor_df
```

```
##   RACACOR       n raca_cor_factor
## 1       1 6794687          branca
## 2       4 5295315           parda
## 3       2 1127508           preta
## 4    <NA>  364106            <NA>
## 5       3   74842         amarela
## 6       5   46782        indígena
## 7       9       5        ignorado
```

5.3 Transformar variável Raça/Cor

Transforma coluna com caracteres em nomes de raça/cor

```
racacor_df <- racacor_df %>%
  dplyr::mutate(raca_cor_factor = ifelse(racacor_df$RACACOR == "1", "branca",
                                         ifelse(racacor_df$RACACOR == "2", "preta",
                                                ifelse(racacor_df$RACACOR == "3", "ama
                                                       ifelse(racacor_df$RACACOR == "4
                                                              ifelse(racacor_df$RACACO
                                                                     ifelse(racacor_df
```

Verificar objeto racacor_df

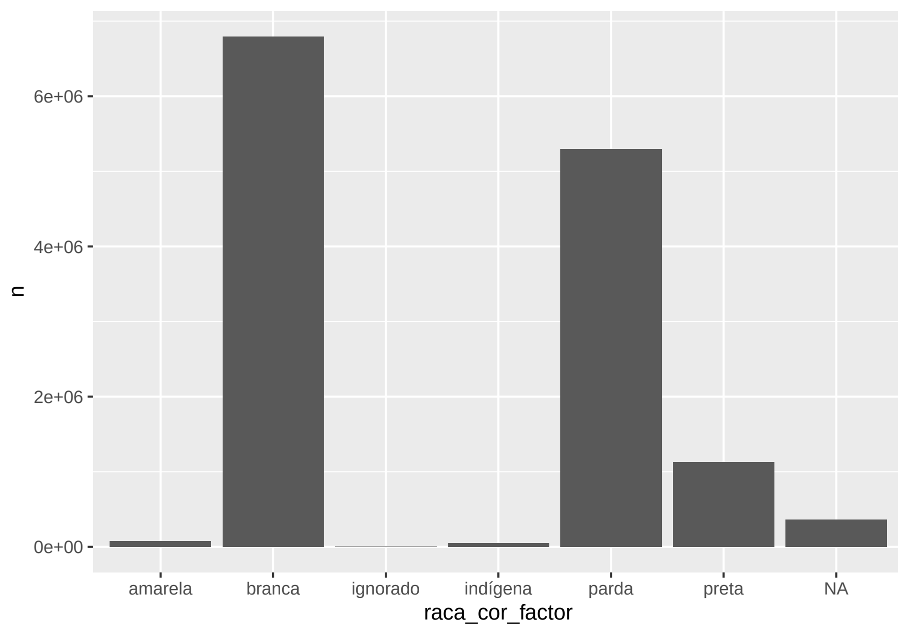
```
racacor_df
```

```
##   RACACOR       n raca_cor_factor
## 1       1 6794687          branca
## 2       4 5295315           parda
## 3       2 1127508           preta
## 4    <NA>  364106            <NA>
## 5       3   74842         amarela
## 6       5   46782        indígena
## 7       9       5        ignorado
```

5.4 Gráfico de barras por categoria da Variável Raça/Cor

Nesta parte, poderemos estimar o número de indivíduos sob risco, considerando as categorias raciais brasileiras.

```
library(ggplot2)
# Basic barplot
ggplot(data=racacor_df, aes(x=raca_cor_factor, y=n)) +
  geom_bar(stat="identity")
```




```
dados_agregados_year <- readRDS("./data/dados_agregados_year.rds")
dados_agregados_year
```

```
## # A tibble: 10 x 2
##    year        n
##    <chr>   <int>
##  1 2015  1264175
```

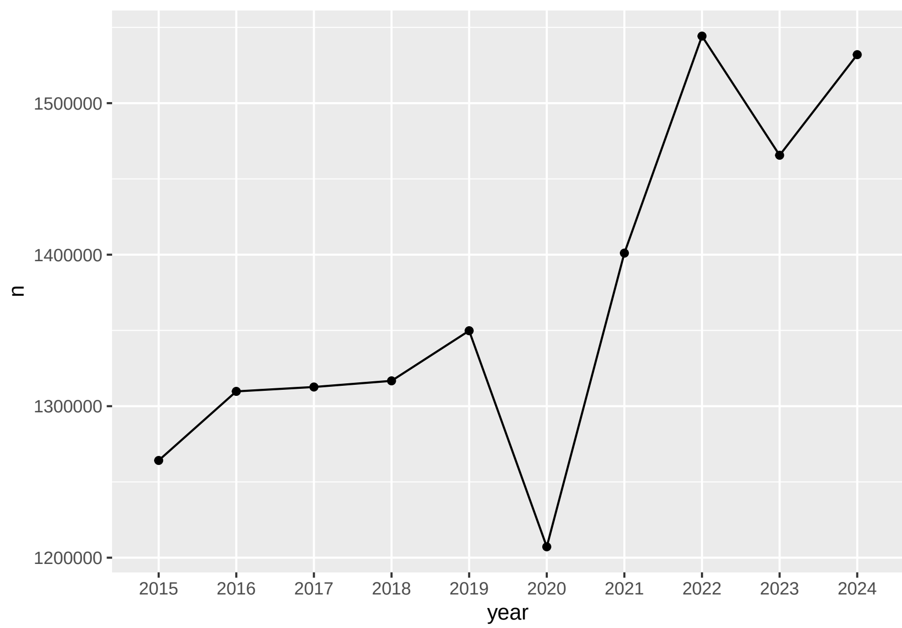
```
##  2 2016  1309774
##  3 2017  1312663
##  4 2018  1316719
##  5 2019  1349801
##  6 2020  1207189
##  7 2021  1401033
##  8 2022  1544266
##  9 2023  1465610
## 10 2024  1532015
```

Remover NAs dos dados agregados por ano

```
dados_agregados_year <- dados_agregados_year[ which(
  dados_agregados_year$year != "NA"), ]
```

5.5 Mortalidade Anual Total

```
library(ggplot2)
# Basic line plot with points
ggplot(data=dados_agregados_year, aes(x=year, y=n, group=1)) +
  geom_line()+
  geom_point()
```



5.5.1 Mortalidade Anual Total Usando Variável Raça/Cor

Mortalidade anual total (figura acima) e mortalidade anual usando a variável Raça/Cor (figura abaixo) foram feitas usando **?** como referência.

Carregar dados_appended_melted

```
dados_appended_melted <- readRDS("./data/dados_appended_melted.rds")
dados_appended_melted
```

```
##    year variable  value
## 1  2015        1 643928
## 2  2016        1 665737
## 3  2017        1 663147
## 4  2018        1 668707
## 5  2019        1 686474
```

```
## 6   2020          1 523848
## 7   2021          1 644210
## 8   2022          1 787363
## 9   2023          1 737590
## 10  2024          1 773683
## 11  2015          2  93830
## 12  2016          2  97747
## 13  2017          2  99745
## 14  2018          2 102889
## 15  2019          2 107306
## 16  2020          2 107171
## 17  2021          2 122852
## 18  2022          2 131005
## 19  2023          2 129047
## 20  2024          2 135916
## 21  2015          3   7051
## 22  2016          3   7276
## 23  2017          3   7375
## 24  2018          3   7524
## 25  2019          3   7359
## 26  2020          3   4716
## 27  2021          3   5313
## 28  2022          3   9261
## 29  2023          3   9053
## 30  2024          3   9914
## 31  2015          4 458807
## 32  2016          4 483026
## 33  2017          4 497839
## 34  2018          4 496106
## 35  2019          4 509936
## 36  2020          4 531808
## 37  2021          4 587323
## 38  2022          4 582469
## 39  2023          4 561438
## 40  2024          4 586563
## 41  2015          5   3640
## 42  2016          5   3867
## 43  2017          5   3911
## 44  2018          5   4261
## 45  2019          5   4273
## 46  2020          5   5173
## 47  2021          5   5361
## 48  2022          5   5343
## 49  2023          5   5398
## 50  2024          5   5555
## 51  2015          9      0
```

```
## 52 2016        9      5
## 53 2017        9      0
## 54 2018        9      0
## 55 2019        9      0
## 56 2020        9      0
## 57 2021        9      0
## 58 2022        9      0
## 59 2023        9      0
## 60 2024        9      0
## 61 2015       NA  56919
## 62 2016       NA  52116
## 63 2017       NA  40646
## 64 2018       NA  37232
## 65 2019       NA  34453
## 66 2020       NA  34473
## 67 2021       NA  35974
## 68 2022       NA  28825
## 69 2023       NA  23084
## 70 2024       NA  20384
```

Obter variável raça/cor como palavras descritivas ao invés de números.

```r
dados_appended_melted <- dados_appended_melted %>%
  dplyr::mutate(raca_cor_factor = ifelse(dados_appended_melted$variable == "1", "branca",
                                         ifelse(dados_appended_melted$variable == "2", "preta",
                                                ifelse(dados_appended_melted$variable == "3", "am
                                                       ifelse(dados_appended_melted$variable == '
                                                              ifelse(dados_appended_melted$variab
                                                                     ifelse(dados_appended_meltec

# Checar dados_appended_melted
dados_appended_melted
```

```
##    year variable  value raca_cor_factor
## 1  2015        1 643928          branca
## 2  2016        1 665737          branca
## 3  2017        1 663147          branca
## 4  2018        1 668707          branca
## 5  2019        1 686474          branca
## 6  2020        1 523848          branca
## 7  2021        1 644210          branca
## 8  2022        1 787363          branca
## 9  2023        1 737590          branca
```

```
## 10 2024         1 773683          branca
## 11 2015         2  93830           preta
## 12 2016         2  97747           preta
## 13 2017         2  99745           preta
## 14 2018         2 102889           preta
## 15 2019         2 107306           preta
## 16 2020         2 107171           preta
## 17 2021         2 122852           preta
## 18 2022         2 131005           preta
## 19 2023         2 129047           preta
## 20 2024         2 135916           preta
## 21 2015         3   7051         amarela
## 22 2016         3   7276         amarela
## 23 2017         3   7375         amarela
## 24 2018         3   7524         amarela
## 25 2019         3   7359         amarela
## 26 2020         3   4716         amarela
## 27 2021         3   5313         amarela
## 28 2022         3   9261         amarela
## 29 2023         3   9053         amarela
## 30 2024         3   9914         amarela
## 31 2015         4 458807           parda
## 32 2016         4 483026           parda
## 33 2017         4 497839           parda
## 34 2018         4 496106           parda
## 35 2019         4 509936           parda
## 36 2020         4 531808           parda
## 37 2021         4 587323           parda
## 38 2022         4 582469           parda
## 39 2023         4 561438           parda
## 40 2024         4 586563           parda
## 41 2015         5   3640        indígena
## 42 2016         5   3867        indígena
## 43 2017         5   3911        indígena
## 44 2018         5   4261        indígena
## 45 2019         5   4273        indígena
## 46 2020         5   5173        indígena
## 47 2021         5   5361        indígena
## 48 2022         5   5343        indígena
## 49 2023         5   5398        indígena
## 50 2024         5   5555        indígena
## 51 2015         9      0        ignorado
## 52 2016         9      5        ignorado
## 53 2017         9      0        ignorado
## 54 2018         9      0        ignorado
## 55 2019         9      0        ignorado
```

```
## 56 2020        9      0        ignorado
## 57 2021        9      0        ignorado
## 58 2022        9      0        ignorado
## 59 2023        9      0        ignorado
## 60 2024        9      0        ignorado
## 61 2015       NA  56919               0
## 62 2016       NA  52116               0
## 63 2017       NA  40646               0
## 64 2018       NA  37232               0
## 65 2019       NA  34453               0
## 66 2020       NA  34473               0
## 67 2021       NA  35974               0
## 68 2022       NA  28825               0
## 69 2023       NA  23084               0
## 70 2024       NA  20384               0
```

Remove NA in year

```
dados_appended_melted <- dados_appended_melted[ which(
  dados_appended_melted$year != "NA"), ]
```

Remove 0 in raça/cor

```
dados_appended_melted <- dados_appended_melted[ which(
  dados_appended_melted$raca_cor_factor != "0"), ]

dados_appended_melted
```

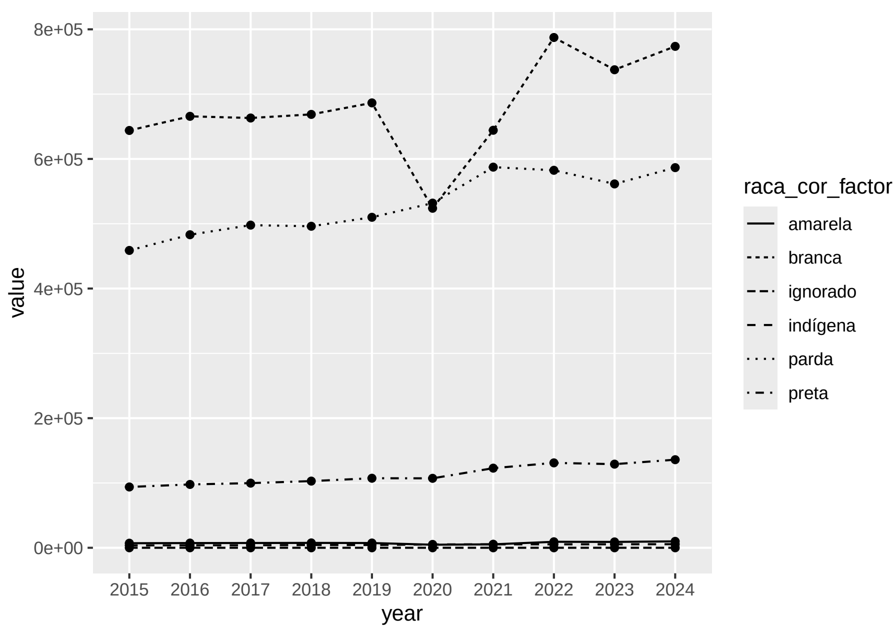
```
##    year variable  value raca_cor_factor
## 1  2015        1 643928          branca
## 2  2016        1 665737          branca
## 3  2017        1 663147          branca
## 4  2018        1 668707          branca
## 5  2019        1 686474          branca
## 6  2020        1 523848          branca
## 7  2021        1 644210          branca
## 8  2022        1 787363          branca
## 9  2023        1 737590          branca
## 10 2024        1 773683          branca
```

```
## 11 2015         2  93830           preta
## 12 2016         2  97747           preta
## 13 2017         2  99745           preta
## 14 2018         2 102889           preta
## 15 2019         2 107306           preta
## 16 2020         2 107171           preta
## 17 2021         2 122852           preta
## 18 2022         2 131005           preta
## 19 2023         2 129047           preta
## 20 2024         2 135916           preta
## 21 2015         3   7051         amarela
## 22 2016         3   7276         amarela
## 23 2017         3   7375         amarela
## 24 2018         3   7524         amarela
## 25 2019         3   7359         amarela
## 26 2020         3   4716         amarela
## 27 2021         3   5313         amarela
## 28 2022         3   9261         amarela
## 29 2023         3   9053         amarela
## 30 2024         3   9914         amarela
## 31 2015         4 458807           parda
## 32 2016         4 483026           parda
## 33 2017         4 497839           parda
## 34 2018         4 496106           parda
## 35 2019         4 509936           parda
## 36 2020         4 531808           parda
## 37 2021         4 587323           parda
## 38 2022         4 582469           parda
## 39 2023         4 561438           parda
## 40 2024         4 586563           parda
## 41 2015         5   3640        indígena
## 42 2016         5   3867        indígena
## 43 2017         5   3911        indígena
## 44 2018         5   4261        indígena
## 45 2019         5   4273        indígena
## 46 2020         5   5173        indígena
## 47 2021         5   5361        indígena
## 48 2022         5   5343        indígena
## 49 2023         5   5398        indígena
## 50 2024         5   5555        indígena
## 51 2015         9      0        ignorado
## 52 2016         9      5        ignorado
## 53 2017         9      0        ignorado
## 54 2018         9      0        ignorado
## 55 2019         9      0        ignorado
## 56 2020         9      0        ignorado
```

```
## 57 2021         9      0         ignorado
## 58 2022         9      0         ignorado
## 59 2023         9      0         ignorado
## 60 2024         9      0         ignorado
```

Visualizar plot

```
ggplot(dados_appended_melted, aes(x=year, y=value, group=raca_cor_factor)) +
  geom_line(aes(linetype=raca_cor_factor))+
  geom_point()
```




```
# wait to change the variable names:

# names(dados_appended_melted)[names(dados_appended_melted) == 'variable'] <- 'RACACOR'
# names(dados_appended_melted)[names(dados_appended_melted) == 'value'] <- 'n'
```

5.6 Mortalidade Anual Total por Neuroblastoma

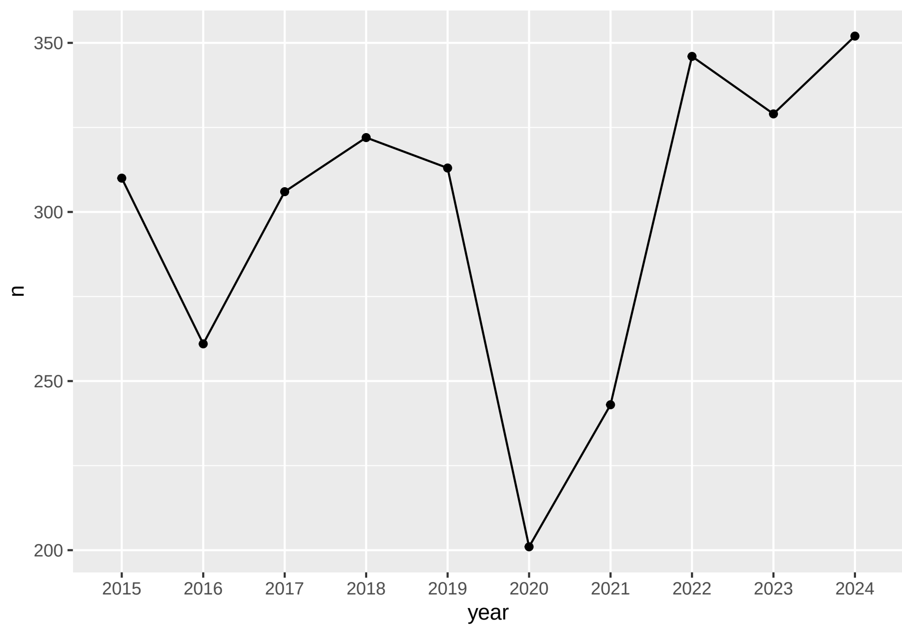
```
dados_nb <- readRDS("./data/dados_nb.rds")

# Checar dados_nb
head(dados_nb)
```

```
##                CONTADOR RACACOR  DTOBITO CAUSABAS   DTNASC IDADE        YMD year
## 876...17551         876       4 20012016     C749 10122010   405 2016-01-20 2016
## 5811...56566     562030       1 14052018     C749 22042018   222 2018-05-14 2018
## 8831...77296     613331       1 11062019     C749 31011964   455 2019-06-11 2019
## 14106...82571    987213       4 01042019     C749 12021950   469 2019-04-01 2019
## 900...87692       78999       1 08032020     C749 02042014   405 2020-03-08 2020
## 8259...95051     470277       1 15022020     C749 21052005   414 2020-02-15 2020
```

```
dados_agregados_nb_year <- dados_nb %>%
  group_by(year) %>%
  summarise(n = n())

# Checar dados_agregados_nb_year
dados_agregados_nb_year
```

```
## # A tibble: 10 x 2
##    year      n
##    <chr> <int>
##  1 2015    310
##  2 2016    261
##  3 2017    306
##  4 2018    322
##  5 2019    313
##  6 2020    201
##  7 2021    243
##  8 2022    346
##  9 2023    329
## 10 2024    352
```

```
ggplot(data=dados_agregados_nb_year, aes(x=year, y=n, group=1)) +
  geom_line()+
  geom_point()
```




```
dados_nb_race_year_selcd <- dados_nb %>%
  select(RACACOR, year)

# Checar dados_nb_race_year_selcd
head(dados_nb_race_year_selcd)
```

```
##                RACACOR year
## 876...17551          4 2016
## 5811...56566         1 2018
## 8831...77296         1 2019
## 14106...82571        4 2019
## 900...87692          1 2020
## 8259...95051         1 2020
```

```
# dados_nb_race_2013_selcd <- dados_estados_nb_2013 %>%
#   select(RACACOR, CAUSABAS)
```

use cast() to convert data frame from long to wide format

```
library(reshape2)
```

```
##
## Attaching package: 'reshape2'

## The following object is masked from 'package:tidyr':
##
##     smiths

## The following objects are masked from 'package:maditr':
##
##     dcast, melt
```

```
wide_dados_nb_appended <- dcast(dados_nb_race_year_selcd, year ~ RACACOR)
```

```
## Using year as value column: use value.var to override.

## Aggregation function missing: defaulting to length
```

Melt the data frame

```
dados_nb_appended_melted <- melt(wide_dados_nb_appended, id.vars = "year")
```

Obter variável raça/cor como palavras descritivas ao invés de números.

```
dados_nb_appended_melted <- dados_nb_appended_melted %>%
  dplyr::mutate(raca_cor_factor = ifelse(dados_nb_appended_melted$variable == "1", "bra
                                        ifelse(dados_nb_appended_melted$variable == "2
                                               ifelse(dados_nb_appended_melted$variabl
                                                      ifelse(dados_nb_appended_melted$
                                                             ifelse(dados_nb_appended_
                                                                    ifelse(dados_nb_ap
```

Remove NA in year

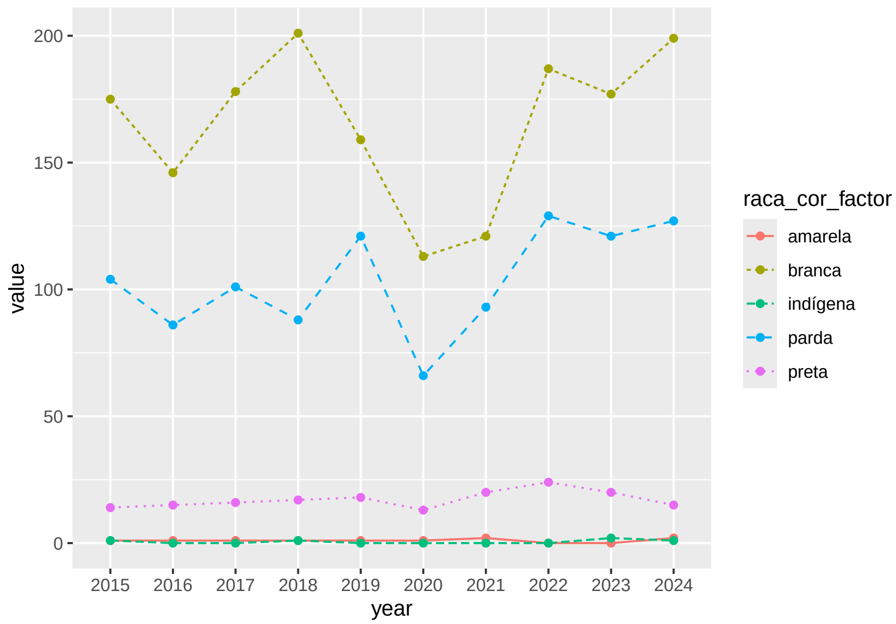
```
dados_nb_appended_melted <- dados_nb_appended_melted[ which(
  dados_nb_appended_melted$year != "NA"), ]
```

Remove 0 in raça/cor

```
dados_nb_appended_melted <- dados_nb_appended_melted[ which(
  dados_nb_appended_melted$raca_cor_factor != "0"), ]
```

Visualizar plot

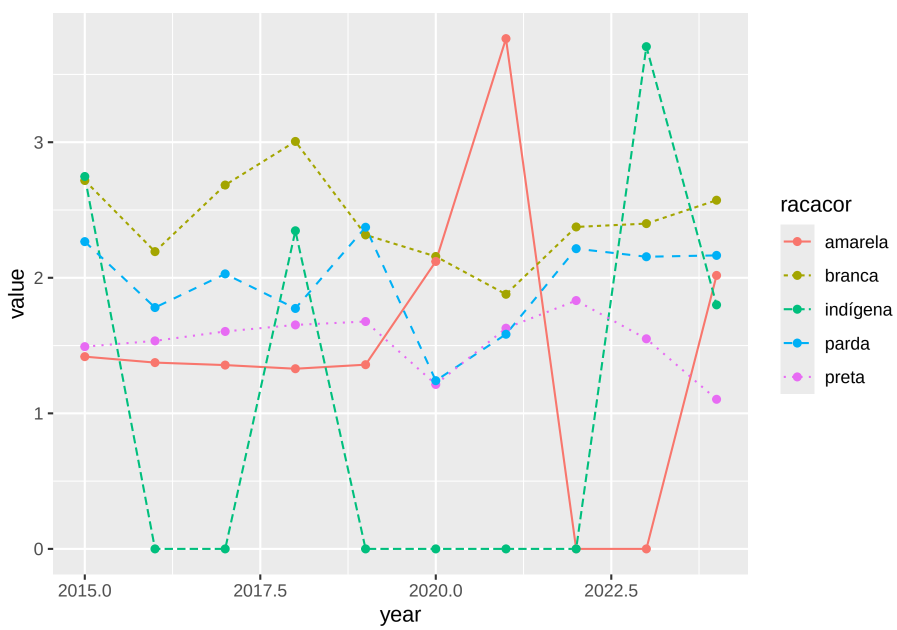
```
ggplot(dados_nb_appended_melted, aes(x=year, y=value, group=raca_cor_factor, color=raca_cor_facto
  geom_line(aes(linetype=raca_cor_factor))+
  geom_point()
```



Visualizar Plot Disparity

```r
library('readxl')
Disparity_coeficient_Brazil_All <- read_excel("./data/Disparity_coeficient_Brazil_All.x
```

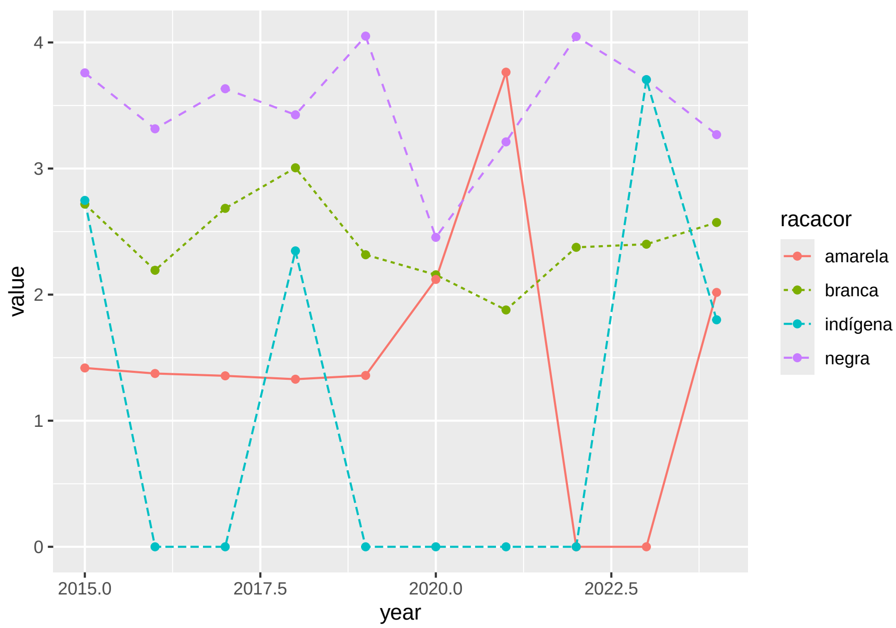
```r
ggplot(Disparity_coeficient_Brazil_All, aes(x=year, y=value, group=racacor, color=racac
  geom_line(aes(linetype=racacor))+
  geom_point()
```



Visualizar Plot Disparity

```r
library('readxl')
Disparity_coeficient_Sum_PardosBlacks_Brazil_All <- read_excel("./data/Disparity_coefic
```

```r
ggplot(Disparity_coeficient_Sum_PardosBlacks_Brazil_All, aes(x=year, y=value, group=rac
  geom_line(aes(linetype=racacor))+
  geom_point()
```



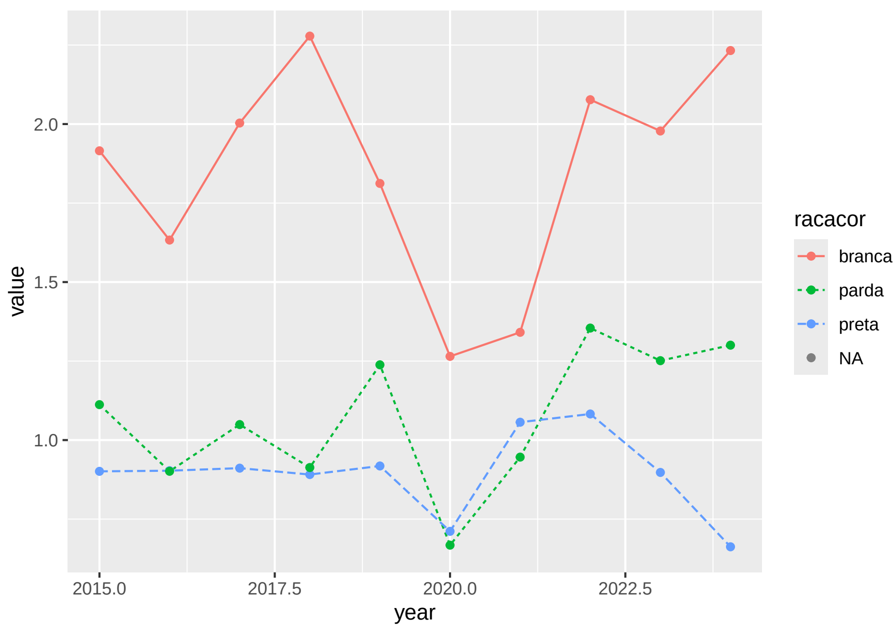
Visualizar Plot Disparity IBGE

```r
library('readxl')
Disparity_ibge_plot <- read_excel("./data/Disparity_ibge_plot.xlsx")
```

```r
ggplot(Disparity_ibge_plot, aes(x=year, y=value, group=racacor, color=racacor)) +
  geom_line(aes(linetype=racacor))+
  geom_point()
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## (`geom_line()`).
```

```
## Warning: Removed 10 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



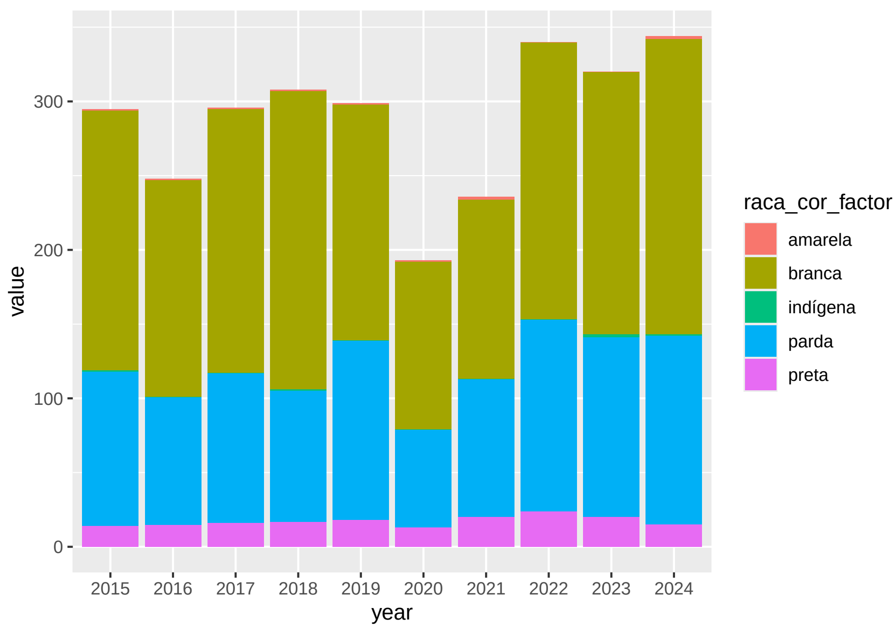
Visualizar Plot Disparity IBGE

```
library('readxl')
Disparity_ibge_negros_plot <- read_excel("./data/Disparity_ibge_negros_plot.xlsx")
```

```
ggplot(Disparity_ibge_negros_plot, aes(x=year, y=value, group=racacor, color=racacor))
  geom_line(aes(linetype=racacor))+
  geom_point()
```

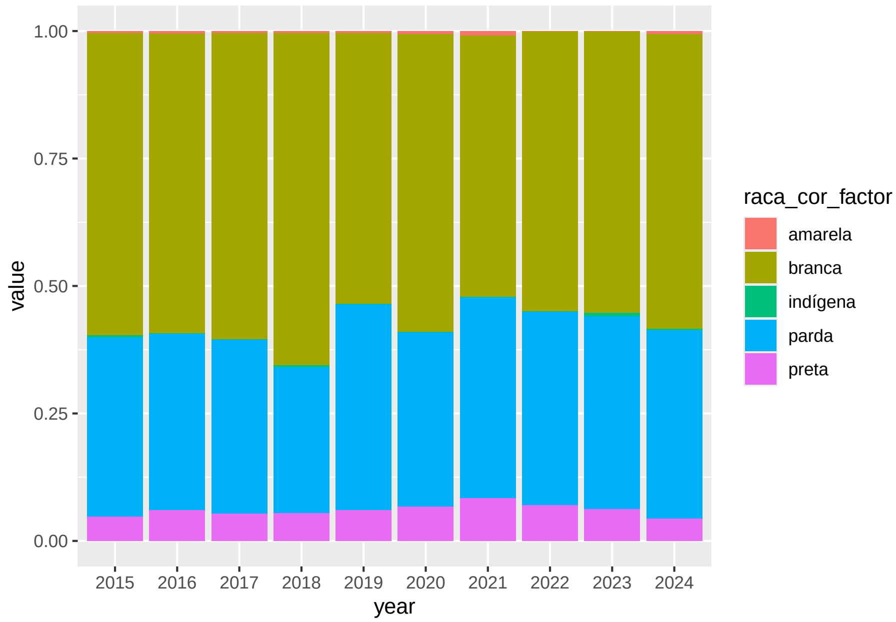
Stacked Graph

```
ggplot(dados_nb_appended_melted, aes(fill=raca_cor_factor, y=value, x=year)) +
    geom_bar(position="stack", stat="identity")
```



Stacked + Graph

```
ggplot(dados_nb_appended_melted, aes(fill=raca_cor_factor, y=value, x=year)) +
    geom_bar(position="fill", stat="identity")
```


1.00
0.75
0.50
0.25
0.00
value
2015
2016
2017
2018
2019
2020
2021
2022
2023
2024
year
raca_cor_factor
amarela
branca
indígena
parda
preta

Chapter 6

Final Words

We have finished a nice book.

Bibliography